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### Display panel and driving method thereof, and display device

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#### Abstract

Provided are a display panel and a driving method thereof, and a display device. A light emission control transistor of a pixel driving circuit in the display panel is connected in series between the anode of a light-emitting element and a first power signal line. A cathode switch transistor is connected in series between the cathode of the light-emitting element and a second power signal line. In at least one light emission stage, the time when the cathode switch transistor is turned on lags behind the time when the light emission control transistor is turned on.

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## Background/Summary

### CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Chinese Patent Application No. 202310344070.8 filed Mar. 29, 2023, the disclosure of which is incorporated herein by reference in its entirety.

### TECHNICAL FIELD

(2) The present disclosure relates to the field of display technologies and, in particular, to a display panel and a driving method thereof, and a display device.

### BACKGROUND

(3) An organic light-emitting diode (OLED) display panel has the advantages of high visibility, high brightness, and being thinner. Therefore, the OLED display panel is more and more widely used.

(4) However, in an existing OLED display panel, there is a problem of display color cast. This affects the display effect of the display panel.

#### SUMMARY

(5) The present disclosure provides a display panel and a driving method thereof, and a display device to improve the color cast problem.

(6) According to one aspect of the present disclosure, a display panel is provided. The display panel includes a cathode switch transistor and a plurality of sub-pixels arranged in an array.

(7) A sub-pixel of the plurality of sub-pixels includes a pixel driving circuit and a light-emitting element electrically connected to the pixel driving circuit. The pixel driving circuit includes a drive transistor and at least one light emission control transistor.

(8) The drive transistor, the light emission control transistor, the at least one light-emitting element, and the cathode switch transistor are connected in series between a first power signal line and a second power signal line. The light emission control transistor is connected in series between the anode of the light-emitting element and the first power signal line. The cathode switch transistor is connected in series between the cathode of the light-emitting element and the second power signal line.

(9) The gate of the light emission control transistor is electrically connected to a first light emission control signal line. The gate of the cathode switch transistor is electrically connected to a second light emission control signal line. In at least one light emission stage, the start time of an effective pulse of a second light emission control signal of the second light emission control signal line lags behind the start time of an effective pulse of a first light emission control signal of the first light emission control signal line.

(10) According to another aspect of the present disclosure, a display device is provided. The display device includes the display panel described in the first aspect.

(11) According to another aspect of the present disclosure, a driving method of a display panel is provided. The display panel includes a cathode switch transistor and a plurality of sub-pixels arranged in an array. A sub-pixel of the plurality of sub-pixels includes a pixel driving circuit and a light-emitting element electrically connected to the pixel driving circuit. The pixel driving circuit includes a drive transistor and at least one light emission control transistor. The drive transistor, the light emission control transistor, the at least one light-emitting element, and the cathode switch transistor are connected in series between a first power signal line and a second power signal line. The light emission control transistor is connected in series between the anode of the light-emitting element and the first power signal line. The cathode switch transistor is connected in series between the cathode of the light-emitting element and the second power signal line. The gate of the light emission control transistor is electrically connected to a first light emission control signal line. The gate of the cathode switch transistor is electrically connected to a second light emission control signal line.

(12) The driving method includes that a first light emission control signal is applied to the first light emission control signal line and a second light emission control signal is applied to the second light emission control signal line, where in at least one light emission stage, the start time of an effective pulse of the second light emission control signal lags behind the start time of an effective pulse of the first light emission control signal.

(13) According to the display panel and the driving method thereof, and the display device provided in the embodiments of the present disclosure, the light emission control transistor is connected in series between the anode of the light-emitting element and the first power signal line, the cathode switch transistor is connected in series between the cathode of the light-emitting element and the second power signal line, the gate of the light emission control transistor is

electrically connected to the first light emission control signal line, the gate of the cathode switch transistor is electrically connected to the second light emission control signal line, and in the light emission stage, the start time of an effective pulse of a second light emission control signal of the second light emission control signal line lags behind the start time of an effective pulse of a first light emission control signal of the first light emission control signal line. Thus, in the light emission stage, the light emission control transistor is first turned on so that the pixel capacitor of the light-emitting element is charged, and after the charging of pixel capacitors of light-emitting elements emitting light of different colors is completed, the cathode switch transistor is turned on. Therefore, the light-emitting elements emitting light of different colors emit light simultaneously when the cathode switch transistor is turned on so that the light-emitting duration and luminescence efficiency of the light-emitting elements emitting light of different colors tend to be consistent in the light emission stage, thereby solving the color cast problem caused by the difference in the pixel capacitors of the light-emitting elements emitting light of different colors.

(14) It is to be understood that the contents described in this part are not intended to identify key or important features of the embodiments of the present disclosure, and are not intended to limit the scope of the present disclosure. Other features of the present disclosure are readily understood through the description hereinafter.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

- (1) To illustrate technical solutions in embodiments of the present disclosure more clearly, drawings used in description of the embodiments are briefly described below. Apparently, the drawings described below merely illustrate part of the embodiments of the present disclosure. Those of ordinary skill in the art may obtain other drawings based on the drawings on the premise that no creative work is done.
- (2) FIG. 1 is a diagram illustrating the structure of a display panel according to an embodiment of the present disclosure.
- (3) FIG. 2 is a sectional structure diagram of FIG. 1 taken along direction A-A'.
- (4) FIG. 3 is a diagram illustrating the structure of a pixel driving circuit according to an embodiment of the present disclosure.
- (5) FIG. 4 is a drive timing diagram of a pixel driving circuit according to an embodiment of the present disclosure.
- (6) FIG. 5 is a diagram illustrating the structure of a pixel driving circuit in the related art.
- (7) FIG. 6 is a drive timing diagram of a pixel driving circuit in the related art.
- (8) FIG. 7 is a potential diagram of an N4 node of a pixel driving circuit in a light emission stage in the related art.
- (9) FIG. 8 is a luminescence efficiency graph of a sub-pixel in the related art.
- (10) FIG. 9 is a diagram illustrating the structure of another display panel according to an embodiment of the present disclosure.
- (11) FIG. 10 is a drive timing diagram of a pixel driving circuit in a first sub-pixel row according to an embodiment of the present disclosure.
- (12) FIG. 11 is a drive timing diagram of a pixel driving circuit in a second sub-pixel row according to an embodiment of the present disclosure.
- (13) FIG. 12 is a diagram illustrating the structure of another display panel according to an embodiment of the present disclosure.
- (14) FIG. 13 is a diagram illustrating the structure of another display panel according to an embodiment of the present disclosure.
- (15) FIG. 14 is a diagram illustrating the structure of another display panel according to an

embodiment of the present disclosure.

(16) FIG. 15 is a diagram illustrating the structure of another display panel according to an embodiment of the present disclosure.

(17) FIG. 16 is a diagram illustrating the structure of another display panel according to an embodiment of the present disclosure.

(18) FIG. 17 is a diagram illustrating the structure of another display panel according to an embodiment of the present disclosure.

(19) FIG. 18 is a diagram illustrating the structure of another display panel according to an embodiment of the present disclosure.

(20) FIG. 19 is a diagram illustrating the structure of another display panel according to an embodiment of the present disclosure.

(21) FIG. 20 is a diagram illustrating the structure of another display panel according to an embodiment of the present disclosure.

(22) FIG. 21 is a diagram illustrating the structure of another display panel according to an embodiment of the present disclosure.

(23) FIG. 22 is a sectional structure diagram of FIG. 21 taken along direction C-C'.

(24) FIG. 23 is a diagram illustrating the structure of another display panel according to an embodiment of the present disclosure.

(25) FIG. 24 is a partial sectional structure diagram of a display panel according to an embodiment of the present disclosure.

(26) FIG. 25 is a partial sectional structure diagram of another display panel according to an embodiment of the present disclosure.

(27) FIG. 26 is a diagram illustrating the structure of a display device according to an embodiment of the present disclosure.

(28) FIG. 27 is a flowchart of a driving method of a display panel according to an embodiment of the present disclosure.

#### DETAILED DESCRIPTION

(29) The technical solutions in the embodiments of the present disclosure are described clearly and completely in conjunction with the drawings in the embodiments of the present disclosure from which the solutions of the present disclosure are better understood by those skilled in the art.

Apparently, the embodiments described below are part, not all, of the embodiments of the present disclosure. Based on the embodiments of the present disclosure, all other embodiments obtained by those of ordinary skill in the art on the premise that no creative work is done are within the scope of the present disclosure.

(30) It is to be noted that the terms “first”, “second”, and the like in the description, claims, and drawings of the present disclosure are used for distinguishing between similar objects and are not necessarily used for describing a particular order or sequence. It is to be understood that the data used in this way is interchangeable where appropriate so that the embodiments of the present disclosure described herein may also be implemented in a sequence not illustrated or described herein. In addition, the terms “including”, “having”, or any other variations thereof described herein are intended to encompass a non-exclusive inclusion. For example, a process, method, system, product, or device that includes a series of steps or units may include not only the expressly listed steps or units but also other steps or units that are not expressly listed or are inherent to such a process, method, system, product, or device.

(31) FIG. 1 is a diagram illustrating the structure of a display panel according to an embodiment of the present disclosure. FIG. 2 is a sectional structure diagram of FIG. 1 taken along direction A-A'. FIG. 3 is a diagram illustrating the structure of a pixel driving circuit according to an embodiment of the present disclosure. FIG. 4 is a drive timing diagram of a pixel driving circuit according to an embodiment of the present disclosure. As shown in FIGS. 1 to 4, a display panel provided in this embodiment of the present disclosure includes a cathode switch transistor M0 and multiple sub-

pixels **11** arranged in an array. A sub-pixel **11** includes a pixel driving circuit **111** and a light-emitting element **112** electrically connected to the pixel driving circuit **111**. The pixel driving circuit **111** includes a drive transistor **M3** and at least one light emission control transistor **M16**. The drive transistor **M3**, the light emission control transistor **M16**, the light-emitting element **112**, and the cathode switch transistor **M0** are connected in series between a first power signal line **PVDD** and a second power signal line **PVEE**. The light emission control transistor **M16** is connected in series between the anode of the light-emitting element **112** and the first power signal line **PVDD**. The cathode switch transistor **M0** is connected in series between the cathode of the light-emitting element **112** and the second power signal line **PVEE**. The gate of the light emission control transistor **M16** is electrically connected to a first light emission control signal line **EM1**. The gate of the cathode switch transistor **M0** is electrically connected to a second light emission control signal line **EM2**. In at least one light emission stage, the start time of an effective pulse of a second light emission control signal of the second light emission control signal line **EM2** lags behind the start time of an effective pulse of a first light emission control signal of the first light emission control signal line **EM1**.

(32) Specifically, as shown in FIGS. **1** to **4**, the display panel provided in this embodiment may be an organic light-emitting diode (OLED) display panel. Multiple sub-pixels **11** arranged in an array are disposed on the display panel. The multiple sub-pixels **11** may include at least two sub-pixels **11** of different colors. For example, as shown in FIG. **1**, the multiple sub-pixels **11** may include a red sub-pixel **R**, a green sub-pixel **G**, and a blue sub-pixel **B**. In some other embodiments, a white sub-pixel is also included to implement color image display.

(33) In FIG. **1**, only a standard RGB arrangement in which multiple sub-pixels **11** are arranged on the display panel is illustrated. In other embodiments, the multiple sub-pixels **11** may be arranged in another arrangement. In addition, the shape of the light-emitting region of a sub-pixel **11** includes, but is not limited to, a rectangle shown in FIG. **1**. In other embodiments, the shape of the light-emitting region of the sub-pixel **11** may also be designed according to actual requirements.

(34) Further, the sub-pixel **11** includes the pixel driving circuit **111** and the light-emitting element **112**. As shown in FIG. **2**, taking the light-emitting element **112** being an organic light-emitting diode as an example for illustration, the light-emitting element **112** may include an anode **21**, a light-emitting layer **22**, and a cathode **23** which are stacked; and when electrons and holes are injected into the light-emitting layer **22** from the cathode **23** and the anode **21**, respectively, excitons are formed in the light-emitting layer **22**, and the light-emitting molecules are excited so that the light-emitting layer **22** emits visible light. Different materials of light-emitting layers **22** are provided to emit visible light of different colors.

(35) With continued reference to FIG. **2** and FIG. **3**, the pixel driving circuit **111** is electrically connected to the light-emitting element **112**. The pixel driving circuit **111** is configured to transmit a light-emitting drive current to the light-emitting element **112** under the action of a signal of a drive signal line (for example, a scan signal line, a data signal line, or a power signal line) on the display panel, thereby providing a drive current for the light-emitting element **112** to drive the light-emitting element **112** to emit light.

(36) Specifically, as shown in FIG. **2** and FIG. **3**, in the pixel driving circuit **111**, the light-emitting element **112**, the drive transistor **M3**, and the light emission control transistor **M16** are connected in series between the first power signal line **PVDD** and the second power signal line **PVEE**, and the light emission control transistor **M16** is connected in series between the anode of the light-emitting element **112** and the first power signal line **PVDD**. The first power signal line **PVDD** is configured to transmit a first power voltage. The second power signal line **PVEE** is configured to transmit a second power voltage. The first power voltage is greater than the second power voltage.

(37) The drive transistor **M3** can be turned on according to the gate potential of the drive transistor **M3**. The drive current formed by the turn-on of the drive transistor **M3** is used for driving the light-emitting element **112** to emit light. The drive transistor **M3** is used as a driving tube. The gate

potential of the drive transistor M3 determines the magnitude of the current of the turn-on of the drive transistor M3. Thus, the brightness of the light-emitting element 112 can be adjusted by controlling the gate voltage of the drive transistor M3 to implement grayscale control.

(38) The light emission control transistor M16 is configured to control turn on or off between the drive transistor M3 and the first power signal terminal PVDD. The light emission control transistor M16 is used as a switch transistor. The light emission control transistor M16 implements turn on and off according to the gate voltage of the light emission control transistor M16. As shown in FIG. 3, the gate of the light emission control transistor M16 is electrically connected to the light emission control signal line EM1 so that the light emission control transistor M16 can be turned on or off under the control of the first light emission control signal transmitted by the light emission control signal line EM1.

(39) FIG. 5 is a diagram illustrating the structure of a pixel driving circuit in the related art. FIG. 6 is a drive timing diagram of a pixel driving circuit in the related art. As shown in FIG. 5 and FIG. 6, exemplarily, taking the pixel driving circuit 111 being a 7T1C circuit as an example, the pixel driving circuit 111 may further include a first reset transistor M5, a data write transistor M2, an additional transistor M4, a light-emitting reset transistor M7, and a first capacitor Cst.

(40) The gate of the first reset transistor M5 is electrically connected to a first scan signal line S1. The first pole of the first reset transistor M5 is electrically connected to a reference signal line Vref. The second pole of the first reset transistor M5 and the gate of the drive transistor M3 are connected to a first node N1.

(41) The gate of the data write transistor M2 is electrically connected to a second scan signal line S2. The first pole of the data write transistor M2 is electrically connected to a data signal line Vdata. The second pole of the data write transistor M2 is electrically connected to the first pole of the drive transistor M3.

(42) The gate of the additional transistor M4 is electrically connected to the second scan signal line S2. The first pole of the additional transistor M4 is electrically connected to the second pole of the drive transistor M3. The second pole of the additional transistor M4 is electrically connected to the first node N1.

(43) The gate of the light-emitting reset transistor M7 is connected to the first scan signal line S1. The first pole of the light-emitting reset transistor M7 is electrically connected to the reference signal line Vref. The second pole of the light-emitting reset transistor M7 is electrically connected to the anode of the light-emitting element 112.

(44) One end of the first capacitor Cst is electrically connected to the first power signal line PVDD. Another end of the first capacitor Cst is electrically connected to the first node N1.

(45) As shown in FIG. 5 and FIG. 6, the driving process of the pixel driving circuit 111 is, for example, as follows.

(46) In an initialization stage T1, the first scan signal Scant on the first scan signal line S1 enables the first reset transistor M5 to be turned on. The reference voltage on the reference signal line Vref is applied to one end of the first capacitor Cst through the first reset transistor M5, that is, the potential of the first node N1 is the reference voltage, to reset the first node N1. At this time, the gate potential of the drive transistor M3 is also the reference voltage.

(47) Meanwhile, in the initialization stage T1, the first scan signal Scant on the first scan signal line S1 enables the light-emitting reset transistor M7 to be turned on. The light-emitting reset transistor M7 writes the reference voltage on the reference signal line Vref into the anode of the light-emitting element 112 and resets the anode potential of the light-emitting element 112. Thus, the influence of the anode voltage of the light-emitting element 112 in the previous frame on the anode voltage of the light-emitting element 112 in the subsequent frame can be reduced, thereby contributing to the improvement of the display uniformity.

(48) In a data signal voltage write stage T2, the second scan signal Scan2 on the second scan signal line S2 enables the data write transistor M2 and the additional transistor M4 to be turned on. At this

time, the gate potential of the drive transistor M3 is the reference voltage. The drive transistor M3 is also turned on. The data signal voltage on the data signal line Vdata is applied to the first node N1 through the data write transistor M2, the drive transistor M3, and the additional transistor M4, thereby writing the data signal voltage into the first capacitor Cst.

(49) In a light emission stage T3, the first light emission control signal Emit1 on the light emission control signal line EM1 enables the light emission control transistor M16 to be turned on. A current path is formed between the first power signal line PVDD, the light emission control transistor M16, the drive transistor M3, the light-emitting element 112, and the second power signal line PVEE. Thus, the drive current generated by the drive transistor M3 is supplied to the light-emitting element 112, thereby driving the light-emitting element 112 to emit light by the drive transistor M3 to implement the light-emitting and display functions of the display panel.

(50) FIG. 7 is a voltage diagram of an N4 node of a pixel driving circuit in a light emission stage in the related art. FIG. 8 is a luminescence efficiency graph of one sub-pixel in the related art. As shown in FIGS. 5 to 8, it is found that the first light emission control signal Emit1 on the light emission control signal line EM1 in the light emission stage T3 enables the light emission control transistor M16 to be turned on. The light-emitting element 112 has a pixel capacitor. After the light emission control transistor M16 is turned on, the pixel capacitor of the light-emitting element 112 is charged first. At this time, the anode potential of the light-emitting element 112 (that is, the potential of the N4 node) gradually rises. When the pixel capacitor of the light-emitting element 112 is fully charged, the light-emitting element 112 starts to emit light continuously.

(51) Since light-emitting elements 112 emitting light of different colors have different materials of light-emitting layers 22, the light-emitting elements 112 emitting light of different colors have different pixel capacitors. A light-emitting element 112 having a smaller pixel capacitor needs a shorter time to charge the pixel capacitor of the light-emitting element 112 in the light emission stage T3. However, a light-emitting element 112 having a large pixel capacitor needs a longer time to charge the pixel capacitor of the light-emitting element 112 in the light emission stage T3. The light-emitting element 112 does not emit light before the pixel capacitor of the light-emitting element 112 is fully charged. Therefore, the longer the charging time of the pixel capacitor of the light-emitting element 112 in the limited duration of the light emission stage T3 within one frame, the later the light-emitting start time of the light-emitting element 112, and the shorter the light-emitting duration of the light-emitting element 112, so the luminescence efficiency of the light-emitting element 112 is lower. Similarly, the shorter the charging time of the pixel capacitor of the light-emitting element 112, the earlier the light-emitting start time of the light-emitting element 112, and the longer the light-emitting duration of the light-emitting element 112, so the luminescence efficiency of the light-emitting element 112 is higher.

(52) Therefore, as shown in FIG. 8, light-emitting elements 112 (such as a red sub-pixel R, a green sub-pixel G, and a blue sub-pixel B in FIG. 8) emitting light of different colors have different charging time (or light-emitting time) of the light-emitting elements 112 to pixel capacitors in the light emission stage T3 due to the difference of the pixel capacitors of the light-emitting elements 112. Thus, the luminescence efficiency changes of the light-emitting elements 112 emitting light of different colors with low grayscale are quite different, thereby causing a color cast problem.

(53) Based on the preceding technical problems, as shown in FIG. 3 and FIG. 4, in this embodiment, the cathode switch transistor M0 is connected in series between the cathode of the light-emitting element 112 and the second power signal line PVEE. The gate of the cathode switch transistor M0 is electrically connected to the second light emission control signal line EM2.

(54) In at least one light emission stage T3 in the duration of one frame, the start time of an effective pulse of a second light emission control signal Emit2 of the second light emission control signal line EM2 lags behind the start time of an effective pulse of a first light emission control signal Emit1 of the first light emission control signal line EM1.

(55) In this manner, as shown in FIG. 3 and FIG. 4, a light emission stage T3 includes a first light-



emitting sub-stage T31 and a second light-emitting sub-stage T32 disposed in sequence. The first light-emitting sub-stage T31 is the time period between the start time of the effective pulse of the second light emission control signal Emit2 of the second light emission control signal line EM2 and the start time of the effective pulse of the first light emission control signal Emit1 of the first light emission control signal line EM1.

(56) With continued reference to FIG. 3 and FIG. 4, in the first light-emitting sub-stage T31, the first light emission control signal Emit1 on the light emission control signal line EM1 is converted into an effective pulse. Thus, the light emission control transistor M16 is turned on. The second light emission control signal Emit2 on the second light emission control signal line EM2 controls the cathode switching transistor M0 to be turned off. At this time, a path between the anode of the light-emitting element 112 and the first power signal line PVDD is conductive. The drive current formed by the turn-on of the drive transistor M3 charges the pixel capacitor of the light-emitting element 112. A path between the cathode of the light-emitting element 112 and the second power signal line PVEE is not conductive. Therefore, in the first light-emitting sub-stage T31, the light-emitting element 112 does not emit light even if the pixel capacitor of the light-emitting element 112 is fully charged.

(57) In the second light-emitting sub-stage T32, the first light emission control signal Emit1 on the light emission control signal line EM1 maintains an effective pulse. Thus, the light emission control transistor M16 maintains on. That is, the path between the anode of the light-emitting element 112 and the first power signal line PVDD maintains conductive. The second light emission control signal Emit2 on the second light emission control signal line EM2 is converted into an effective pulse so that the cathode switch transistor M0 is turned on. At this time, at the start time of the effective pulse of the second light emission control signal Emit2 of the second light emission control signal line EM2, the cathode switch transistor M0 is turned on. Thus, the path between the cathode of the light-emitting element 112 and the second power signal line PVEE is conductive. Pixel capacitors of light-emitting elements 112 emitting light of different colors have completed the charging process in the first light-emitting sub-stage T31, so that the light-emitting elements 112 emitting light of different colors can directly emit light when the cathode switch transistor M0 is turned on without having to go through the charging stage of the pixel capacitors. Thus, the light-emitting elements 112 emitting light of different colors start to emit light simultaneously in the light emission stage T3. That is, the light-emitting start time of the light-emitting elements 112 emitting light of different colors becomes consistent in the light emission stage T3, and thus the light-emitting duration of the light-emitting elements 112 emitting light of different colors becomes more consistent in the light emission stage T3. Moreover, the luminescence efficiency of the light-emitting elements 112 tends to be consistent, thereby solving the color cast problem caused by the difference in the pixel capacitors of the light-emitting elements 112 emitting light of different colors.

(58) It is to be noted that the pixel driving circuit shown in FIG. 3 adds the cathode switch transistor M0 based on the 7T1C circuit shown in FIG. 5. However, the application of the cathode switch transistor M0 is not limited to the pixel driving circuit structures shown in FIG. 3 and FIG. 5. In other types of pixel driving circuit structures, the cathode switch transistor M0 may be connected in series to the cathode of a light-emitting element 112 so as to achieve that light-emitting elements 112 emitting light of different colors emit light simultaneously when the cathode switch transistor M0 is turned on, and thus the light-emitting duration and the luminescence efficiency of the light-emitting elements 112 emitting light of different colors tend to be consistent in the light emission stage T3, thereby solving the color cast problem caused by the difference in the pixel capacitors of the light-emitting elements 112 emitting light of different colors.

(59) With continued reference to FIG. 3 and FIG. 5, optionally, at least one light emission control transistor M16 includes a first light emission control transistor M1 and a second light emission control transistor M6. The gate of the first light emission control transistor M1 is electrically

connected to the first light emission control signal line EM1. The first pole of the first light emission control transistor M1 is electrically connected to the first power signal line PVDD. The second pole of the first light emission control transistor M1 is electrically connected to the first pole of the drive transistor M3. The gate of the second light emission control transistor M6 is electrically connected to the first light emission control signal line EM1. The first pole of the second light emission control transistor M6 is electrically connected to the second pole of the drive transistor M3. The second pole of the second light emission control transistor M6 is electrically connected to the light-emitting element 112.

(60) The light emission control transistor M16 is set to include the first light emission control transistor M1 and the second light emission control transistor M6 so that in the non-light emission stage, the first light emission control transistor M1 and the second light emission control transistor M6 are cut off simultaneously under the action of the first light emission control signal Emit1 on the light emission control signal line EM1 to ensure the disconnection between the light-emitting element 112 and the first power signal line PVDD. Thus, it can be avoided that the light emission control transistor M16 generates the leakage current to cause the light-emitting element 112 to emit undesired light in the non-light emission stage to influence the display effect.

(61) In summary, according to the display panel in this embodiment of the present disclosure, the light emission control transistor is connected in series between the anode of the light-emitting element and the first power signal line, and the cathode switch transistor is connected in series between the cathode of the light-emitting element and the second power signal line, the gate of the light emission control transistor is electrically connected to the first light emission control signal line, the gate of the cathode switch transistor is electrically connected to the second light emission control signal line, and in the light emission stage, the start time of the effective pulse of the second light emission control signal of the second light emission control signal line lags behind the start time of the effective pulse of the first light emission control signal of the first light emission control signal line. Thus, in the light emission stage, the light emission control transistor is first turned on so that the pixel capacitor of the light-emitting element is charged, and after the charging of pixel capacitors of light-emitting elements emitting light of different colors is completed, the cathode switch transistor is turned on. Therefore, the light-emitting elements emitting light of different colors emit light simultaneously when the cathode switch transistor is turned on so that the light-emitting duration and luminescence efficiency of the light-emitting elements emitting light of different colors tend to be consistent in the light emission stage, thereby solving the color cast problem caused by the difference in the pixel capacitors of the light-emitting elements emitting light of different colors.

(62) With continued reference to FIGS. 1 to 4, optionally, emitted colors of at least part of sub-pixels 11 are different. Light-emitting elements 112 in the sub-pixels 11 have corresponding pixel capacitors. Capacitance values of the pixel capacitors of the light-emitting elements 112 in the sub-pixels 11 of different emitted colors are different. In at least one row of sub-pixels, a sub-pixel with the maximum capacitance value of the pixel capacitor of a light-emitting element 112 is the maximum capacitance sub-pixel. The charging time of the pixel capacitor of the light-emitting element 112 in the maximum capacitance sub-pixel is  $t1$ . For at least one row of sub-pixels 11, in the light emission stage T3, a duration in which the start time of an effective pulse of a second light emission control signal Emit2 of a second light emission control signal line EM2 corresponding to a sub-pixel 11 lags behind the start time of an effective pulse of a first light emission control signal Emit1 of a first light emission control signal line EM1 corresponding to the sub-pixel 11 is delay time. The delay time is  $\Delta t$ , and  $\Delta t \geq t1$ .

(63) As described above, the display panel includes sub-pixels 11 of at least two different colors. As shown in FIG. 1, the display panel may include a red sub-pixel R, a green sub-pixel G, and a blue sub-pixel B. In some other embodiments, a white sub-pixel may further be included to implement color image display.

(64) Pixel capacitors of light-emitting elements **112** emitting light of different colors have different capacitance values due to different materials of light-emitting layers **22** of the light-emitting elements **112**.

(65) With continued reference to FIGS. **1** to **4**, the display panel includes multiple data signal lines **Vdata** and multiple second scan signal lines **S2**, the extension directions of the data signal lines **Vdata** and the second scan signal lines **S2** intersect, and the multiple data signal lines **Vdata** and second scan signal lines **S2** intersect to define multiple sub-pixels **11**.

(66) The multiple second scan signal lines **S2** may extend along the row direction and may be arranged along the column direction. One second scan signal line **S2** is disposed correspondingly and electrically connected to pixel driving circuits **111** of one row of sub-pixels **11**. The multiple data signal lines **Vdata** may extend along the column direction and may be arranged along the row direction. One data signal line **Vdata** is disposed correspondingly and connected to pixel driving circuits **111** of one column of sub-pixels **11**. A second scan signal line **S2** is configured to provide a second scan signal **Scan2** for pixel driving circuits **111** corresponding to one row of sub-pixels **11**. Thus, the data signal on the data signal line **Vdata** can be written into the pixel driving circuits **111** corresponding to the one row of sub-pixels **11**. The second scan signal **Scan2** described herein refers to an effective pulse signal output from the second scan signal line **S2**. For example, in the structure of a pixel driving circuit **111** shown in FIG. **3**, the second scan signal **Scan2** output from the second scan signal line **S2** is a signal capable of turning on the data write transistor **M2** and the additional transistor **M4**. The second scan signal **Scan2** is at a low level when the data write transistor **M2** and the additional transistor **M4** are positive channel metal oxide semiconductor (PMOS) transistors. The second scan signal **Scan2** is at a high level when the data write transistor **M2** and the additional transistor **M4** are negative channel metal oxide semiconductor (NMOS) transistors.

(67) Similarly, the first scan signal line **S1**, the light emission control signal line **EM1**, and the second light emission control signal line **EM2** are disposed in a manner similar to the second scan signal line **S2**. That is, one signal line is disposed correspondingly and electrically connected to pixel driving circuits **111** of one row of sub-pixels **11**, thereby providing a corresponding scan signal for the pixel driving circuits **111** in correspondence with the one row of sub-pixels **11**.

(68) Therefore, driving processes of pixel driving circuits **111** of the same row of sub-pixels **11** are performed simultaneously. That is, initialization stages **T1**, data signal voltage write stages **T2**, and light emission stages **T3** (including first light-emitting sub-stages **T31** and second light-emitting sub-stages **T32**) of the same row of sub-pixels **11** are performed simultaneously.

(69) With continued reference to FIG. **1**, in one row of sub-pixels **111**, there may be sub-pixels **11** of at least two different colors, and capacitance values of pixel capacitors of light-emitting elements **112** in the sub-pixels **11** of different colors are different. In one row of sub-pixels **111**, a sub-pixel **11** with the maximum capacitance value of the pixel capacitor of a light-emitting element **112** is the maximum capacitance sub-pixel in the one row of sub-pixels **111**. As described above, the larger the capacitance value of the pixel capacitor of the light-emitting element **112**, the longer the charging time of the pixel capacitor of the light-emitting element **112** in the light emission stage **T3**. Therefore, the sub-pixel **11** with the maximum capacitance value of the pixel capacitor of the light-emitting element **112** is the sub-pixel **11** with the longest charging time required in the light emission stage **T3**.

(70) In this embodiment, the charging time of the pixel capacitor of the light-emitting element **112** in the maximum capacitance sub-pixel is  $t_1$ . For any row of sub-pixels **11**, in the light emission stage **T3**, a duration in which the start time of an effective pulse of a second light emission control signal **Emit2** of a second light emission control signal line **EM2** corresponding to a sub-pixel **11** lags behind the start time of an effective pulse of a first light emission control signal **Emit1** of a first light emission control signal line **EM1** corresponding to the sub-pixel **11** is delay time  $\Delta t$ . As shown in FIG. **4**, the delay time  $\Delta t$  is the duration of the first light-emitting sub-stage **T31**.

(71) By setting the delay time  $\Delta t$  to be greater than or equal to the charging time  $t_1$  of the pixel capacitor of the light-emitting element **112** in the maximum capacitance sub-pixel, it can be ensured that the charging of the pixel capacitor of the light-emitting element **112** in the maximum capacitance sub-pixel can be completed in the first light-emitting sub-stage **T31**. It is to be understood that one a row of sub-pixels **111**, the pixel capacitance of the light-emitting element **112** in the maximum capacitance sub-pixel is maximum, and the charging time required in the light emission stage **T3** is maximum, so that in the one row of sub-pixels **111**, the charging time required for pixel capacitors of light-emitting elements **112** in sub-pixels **111** other than the maximum capacitance sub-pixel is each less than the charging time required for the pixel capacitor of the light-emitting element **112** in the maximum capacitance sub-pixel. Therefore, for the one row of sub-pixels **111**, in the first light-emitting sub-stage **T31**, if the charging of the pixel capacitor of the light-emitting element **112** in the maximum capacitance sub-pixel can be completed, sub-pixels **111** in the one row of sub-pixels **111** other than the maximum capacitance sub-pixel can certainly complete the charging processes of the pixel capacitors in the light-emitting elements **112** in the first light-emitting sub-stage **T31**.

(72) Therefore, in this embodiment, the delay time  $\Delta t$  is set to be greater than or equal to the charging time  $t_1$  of the pixel capacitor of the light-emitting element **112** in the maximum capacitance sub-pixel for any row of sub-pixels **11**, so that for any row of sub-pixels **11**, the charging process of pixel capacitors of light-emitting elements **112** of all sub-pixels **11** in the row of sub-pixels **11** can be completed in the first light-emitting sub-stage **T31**, thereby ensuring that each row of sub-pixels **11** emits light simultaneously when a cathode switch transistor **M0** correspondingly connected to the each row of sub-pixels **11** is turned on (that is, at the start time of the effective pulse of the second light emission control signal **Emit2** of the second light emission control signal line **EM2** or the start time of the second light-emitting sub-stage **T32**) without charging the pixel capacitors of the light-emitting elements **112** again. Therefore, the light-emitting start time of the light-emitting elements **112** emitting light of different colors in the same row of sub-pixels **11** is consistent in the light emission stage **T3**, and thus the light-emitting duration of the light-emitting elements **112** emitting light of different colors tends to be consistent in the light emission stage **T3**. Moreover, the luminescence efficiency of the light-emitting elements **112** tends to be consistent, thereby solving the color cast problem caused by the difference in the pixel capacitors of the light-emitting elements **112** emitting light of different colors.

(73) With continued reference to FIGS. **1** to **4**, optionally, multiple sub-pixels **11** include a first color sub-pixel **11a**, a second color sub-pixel **11b**, and a third color sub-pixel **11c**. A capacitance value of a pixel capacitor of a light-emitting element **112** in the first color sub-pixel **11a** is greater than a capacitance value of a pixel capacitor of a light-emitting element **112** in the second color sub-pixel **11b**. The capacitance value of the pixel capacitor of the light-emitting element **112** in the second color sub-pixel **11b** is greater than a capacitance value of a pixel capacitor of a light-emitting element **112** in the third color sub-pixel **11c**. The charging time of the pixel capacitor of the light-emitting element **112** in the first color sub-pixel **11a** is  $t_2$ , and  $\Delta t \geq t_2$ .

(74) Specifically, as shown in FIG. **1**, the display panel includes three types of sub-pixels **11** emitting light of different colors, respectively the first color sub-pixel **11a**, the second color sub-pixel **11b**, and the third color sub-pixel **11c**, to implement color image display.

(75) The first color sub-pixel **11a** is a sub-pixel **11** having the maximum capacitance value of the pixel capacitor of the light-emitting element **112** among the three types of sub-pixels **11** emitting light of different colors. That is, the first color sub-pixel **11a** is a sub-pixel **11** having the maximum capacitance value of the pixel capacitor of the light-emitting element **112** in the entire display panel. Moreover, the first color sub-pixel **11a** is a sub-pixel **11** which requires the longest charging time in the light emission stage **T3** in the entire display panel.

(76) In this embodiment, the delay time  $\Delta t$  is set to be greater than or equal to the charging time  $t_2$  of the pixel capacitor of the light-emitting element **112** in the first color sub-pixel **11a**, so that it can

be ensured that each row of sub-pixels **11** completes the charging processes of the pixel capacitors of the light-emitting elements **112** in the each row of sub-pixels **11** in the first light-emitting sub-stage **T31** regardless of sub-pixels **11** emitting light of which colors are included in each row of sub-pixels **11**, thereby ensuring that each row of sub-pixels **11** emits light simultaneously when the cathode switch transistor **M0** correspondingly connected to the each row of sub-pixels **11** is turned on (that is, at the start time of the effective pulse of the second light emission control signal **Emit2** of the second light emission control signal line **EM2** or the start time of the second light-emitting sub-stage **T32**) without charging the pixel capacitors of the light-emitting elements **112** again. Thus, the light-emitting start time of the light-emitting elements **112** emitting light of different colors in the same row of sub-pixels **11** is consistent in the light emission stage **T3**, and the light-emitting duration of the light-emitting elements **112** emitting light of different colors tends to be consistent in the light emission stage **T3**. Moreover, the luminescence efficiency of the light-emitting elements **112** tends to be consistent, thereby solving the color cast problem caused by the difference in the pixel capacitors of the light-emitting elements **112** emitting light of different colors.

(77) Optionally, the delay time  $\Delta t$  corresponding to each row of sub-pixels **11** is equal.

(78) As described above, when the delay time  $\Delta t$  is greater than or equal to the charging time  $t_2$  of the pixel capacitor of the light-emitting element **112** in the first color sub-pixel **11a**, it can be ensured that each row of sub-pixels **11** completes the charging processes of the pixel capacitors of the light-emitting elements **112** in the each row of sub-pixels **11** in the first light-emitting sub-stage **T31** regardless of sub-pixels **11** emitting light of which colors are included in each row of sub-pixels **11**.

(79) On this basis, the delay time  $\Delta t$  corresponding to each row of sub-pixels **11** is set to be equal, that is, the duration of the first light-emitting sub-stage **T31** of each row of sub-pixels **11** in the driving process is equal. Thus, driving processes of pixel driving circuits **111** in each row of sub-pixels **11** are consistent.

(80) At this time, as shown in FIG. 4, the second light emission control signal **Emit2** of the second light emission control signal line **EM2** may be a periodic pulse signal having the same waveform as the first light emission control signal **Emit1** of the first light emission control signal line **EM1**. The second light emission control signal **Emit2** may be provided for the second light emission control signal line **EM2** through only one scan driving circuit. The scan driving circuit providing the second light emission control signal **Emit2** may have the same circuit structure as the scan driving circuit providing the first light emission control signal **Emit1**. The output first light emission control signal **Emit1** and second light emission control signal **Emit2** are misaligned only in time. Thus, the design difficulty of the scan driving circuit providing the second light emission control signal **Emit2** can be reduced, making it easy to implement.

Optionally,  $2\ \mu\text{s} \leq \Delta t \leq 4\ \mu\text{s}$ .

(81) It is found that when the delay time  $\Delta t$  corresponding to each row of sub-pixels **11** is greater than or equal to  $2\ \mu\text{s}$ , it can be ensured that each row of sub-pixels **11** completes the charging processes of the pixel capacitors of the light-emitting elements **112** in the each row of sub-pixels **11** in the first light-emitting sub-stage **T31** regardless of sub-pixels **11** emitting light of which colors are included in each row of sub-pixels **11**, thereby ensuring that each row of sub-pixels **11** emits light simultaneously when the cathode switch transistor **M0** correspondingly connected to the each row of sub-pixels **11** is turned on without charging the pixel capacitors of the light-emitting elements **112** again. Thus, the light-emitting start time of the light-emitting elements **112** emitting light of different colors in the same row of sub-pixels **11** is consistent in the light emission stage **T3**, and the light-emitting duration of the light-emitting elements **112** tends to be consistent in the light emission stage **T3**. Moreover, the luminescence efficiency of the light-emitting elements **112** tends to be consistent, thereby solving the color cast problem caused by the difference in the pixel capacitors of the light-emitting elements **112** emitting light of different colors.

(82) Meanwhile, the delay time  $\Delta t$  corresponding to each row of sub-pixels **11** is set to be less than or equal to 4  $\mu s$ . Thus, the duration of the first light-emitting sub-stage **T31** is not too long to severely compress the duration (that is, the duration of the second light-emitting sub-stage **T32**) in which the light-emitting element **112** emits light in the light emission stage **T3**. Moreover, it can be ensured that the light-emitting element **112** has relatively long light-emitting time in the light emission stage **T3**, and the light-emitting element **112** has a relatively high luminescence efficiency.

(83) FIG. 9 is a diagram illustrating the structure of another display panel according to an embodiment of the present disclosure. FIG. 10 is a drive timing diagram of a pixel driving circuit in a first sub-pixel row according to an embodiment of the present disclosure. FIG. 11 is a drive timing diagram of a pixel driving circuit in a second sub-pixel row according to an embodiment of the present disclosure. Optionally, as shown in FIGS. 9 to 11, multiple sub-pixels **11** include first sub-pixel rows **31** and second sub-pixel rows **32**. The maximum capacitance sub-pixel in a first sub-pixel row **31** and the maximum capacitance sub-pixel in a second sub-pixel row **32** have different emitted colors. The capacitance value of the pixel capacitor of the maximum capacitance sub-pixel in the first sub-pixel row **31** is greater than the capacitance value of the pixel capacitor of the maximum capacitance sub-pixel in the second sub-pixel row **32**. The delay time  $\Delta t$  corresponding to the first sub-pixel row **31** is greater than the delay time  $\Delta t$  corresponding to the second sub-pixel row **32**.

(84) The display panel includes the first sub-pixel rows **31** and the second sub-pixel rows **32**. Emitted colors of sub-pixels **11** included in the first sub-pixel row **31** are at least partially different from emitted colors of sub-pixels **11** included in the second sub-pixel row **32**. This causes that the maximum capacitance sub-pixel in the first sub-pixel row **31** and the maximum capacitance sub-pixel in the second sub-pixel row **32** have different emitted colors.

(85) Exemplarily, as shown in FIG. 9, the display panel includes first color sub-pixels **11a**, second color sub-pixels **11b**, and third color sub-pixels **11c**, the capacitance value of the pixel capacitor of the light-emitting element **112** in a first color sub-pixel **11a** is greater than the capacitance value of the pixel capacitor of the light-emitting element **112** in a second color sub-pixel **11b**, and the capacitance value of the pixel capacitor of the light-emitting element **112** in the second color sub-pixel **11b** is greater than the capacitance value of the pixel capacitor of the light-emitting element **112** in a third color sub-pixel **11c**. The first sub-pixel row **31** is formed by arranging first color sub-pixels **11a** and second color sub-pixels **11b**. The second sub-pixel row **32** is formed by arranging second color sub-pixels **11b** and third color sub-pixels **11c**. The maximum capacitance sub-pixel in the first sub-pixel row **31** is the first color sub-pixel **11a**. The maximum capacitance sub-pixel in the second sub-pixel row **32** is the second color sub-pixel **11b**. In this case, the capacitance value of the pixel capacitor of the maximum capacitance sub-pixel (for example, the first color sub-pixel **11a** in FIG. 9) in the first sub-pixel row **31** is greater than the capacitance value of the pixel capacitor of the maximum capacitance sub-pixel (for example, the second color sub-pixel **11b** in FIG. 9) in the second sub-pixel row **32**.

(86) It is to be understood that in order for the first sub-pixel row **31** to complete the charging process of pixel capacitors of light-emitting elements **112** in all sub-pixels **11** in the row in the first light-emitting sub-stage **T31**, it is necessary to set the delay time  $\Delta t$  corresponding to the first sub-pixel row **31** to be greater than or equal to the charging time (for example, the charging time required for the pixel capacitor of the light-emitting element **112** in the first color sub-pixel **11a** in FIG. 9) required for the pixel capacitor of the light-emitting element **112** of the maximum capacitance sub-pixel in the first sub-pixel row **31**, and set the delay time  $\Delta t$  corresponding to the second sub-pixel row **32** to be greater than or equal to the charging time (for example, the charging time required for the pixel capacitor of the light-emitting element **112** in the second color sub-pixel **11b** in FIG. 9) required for the pixel capacitor of the light-emitting element **112** of the maximum capacitance sub-pixel in the second sub-pixel row **32**.

(87) The capacitance value of the pixel capacitor of the maximum capacitance sub-pixel (for

example, the first color sub-pixel **11a** in FIG. **9**) in the first sub-pixel row **31** is greater than the capacitance value of the pixel capacitor of the maximum capacitance sub-pixel (for example, the second color sub-pixel **11b** in FIG. **9**) in the second sub-pixel row **32**, and thus, the charging time (for example, the charging time required for the pixel capacitor of the light-emitting element **112** in the first color sub-pixel **11a** in FIG. **9**) required for the pixel capacitor of the light-emitting element **112** of the maximum capacitance sub-pixel in the first sub-pixel row **31** is greater than the charging time (for example, the charging time required for the pixel capacitor of the light-emitting element **112** in the second color sub-pixel **11b** in FIG. **9**) required for the pixel capacitor of the light-emitting element **112** of the maximum capacitance sub-pixel in the second sub-pixel row **32**.

(88) Therefore, in this embodiment, as shown in FIG. **10** and FIG. **11**, the delay time  $\Delta t$  corresponding to the first sub-pixel row **31** is set to be large to ensure that the delay time  $\Delta t$  corresponding to the first sub-pixel row **31** is greater than or equal to the charging time (for example, the charging time required for the pixel capacitor of the light-emitting element **112** in the first color sub-pixel **11a** in FIG. **9**) required for the pixel capacitor of the light-emitting element **112** of the maximum capacitance sub-pixel in the first sub-pixel row **31**. Thus, it can ensure that the first sub-pixel row **31** completes the charging processes of pixel capacitors of light-emitting elements **112** in sub-pixels **11** in the row in the first light-emitting sub-stage **T31** so that the sub-pixels **11** in the first sub-pixel row **31** can emit light simultaneously when the cathode switch transistor **M0** correspondingly connected to the sub-pixels **11** is turned on.

(89) Meanwhile, the delay time  $\Delta t$  corresponding to the second sub-pixel row **32** is set to be small. When it is ensured that the delay time  $\Delta t$  corresponding to the second sub-pixel row **32** is greater than or equal to the charging time (for example, the charging time required for the pixel capacitor of the light-emitting element **112** in the second color sub-pixel **11b** in FIG. **9**) required for the pixel capacitor of the light-emitting element **112** of the maximum capacitance sub-pixel in the second sub-pixel row **32**, the light-emitting duration (that is, the duration of the second light-emitting sub-stage **T32**) of the light-emitting element **112** of the sub-pixel **11** in the second sub-pixel row **32** in the light emission stage **T3** is greater than the light-emitting duration (that is, the duration of the second light-emitting sub-stage **T32**) of the light-emitting element **112** of the sub-pixel **11** in the first sub-pixel row **31** in the light emission stage **T3**. Thus, the luminescence efficiency of sub-pixels **11** in the second sub-pixel row **32** can be improved while the color cast problem caused by the difference in pixel capacitors of light-emitting elements **112** emitting light of different colors is solved. Moreover, a smaller working current density can be adopted under the condition of implementing the same brightness, which is helpful to prolong the service life of sub-pixels **11** in the second sub-pixel row **32**.

(90) FIG. **12** is a diagram illustrating the structure of another display panel according to an embodiment of the present disclosure. As shown in FIGS. **9** to **12**, optionally, first sub-pixel rows **31** and second sub-pixel rows **32** are alternately arranged along a column direction. Multiple sub-pixels **11** include first color sub-pixels **11a**, second color sub-pixels **11b**, and third color sub-pixels **11c**. The capacitance value of the pixel capacitor of the light-emitting element **112** in a first color sub-pixel **11a** is greater than the capacitance value of the pixel capacitor of the light-emitting element **112** in a second color sub-pixel **11b**. The capacitance value of the pixel capacitor of the light-emitting element **112** in the second color sub-pixel **11b** is greater than the capacitance value of the pixel capacitor of the light-emitting element **112** in a third color sub-pixel **11c**. A first sub-pixel row **31** includes first color sub-pixels **11a** and second color sub-pixels **11b**. A second sub-pixel row **32** includes second color sub-pixels **11b** and third color sub-pixels **11c**. Alternatively, a first sub-pixel row **31** includes first color sub-pixels **11a** and third color sub-pixels **11c**. A second sub-pixel row **32** includes second color sub-pixels **11b** and third color sub-pixels **11c**. The delay time  $\Delta t$  corresponding to the first sub-pixel row **31** is greater than or equal to the charging time of the pixel capacitor of the light-emitting element **112** in a first color sub-pixel **11a**. The delay time  $\Delta t$  corresponding to the second sub-pixel row **32** is greater than or equal to the charging time of the

pixel capacitor of the light-emitting element **112** in a second color sub-pixel **11b**.

(91) Exemplarily, as shown in FIG. 9, first sub-pixel rows **31** and second sub-pixel rows **32** are alternately arranged along the column direction. A first sub-pixel row **31** may be formed by alternately arranging first color sub-pixels **11a** and second color sub-pixels **11b** along the row direction. A second sub-pixel row **32** may be formed by alternately arranging second color sub-pixels **11b** and third color sub-pixels **11c** along the row direction. Moreover, two adjacent sub-pixels **11** along the column direction have different emitted colors. For example, in FIG. 9, second color sub-pixels **11b** and third color sub-pixels **11c** are alternately arranged along the column direction. First color sub-pixels **11a** and second color sub-pixels **11b** are alternately arranged along the column direction.

(92) In this case, as shown in FIG. 9, a  $2 \times 2$  sub-pixel matrix formed by two second color sub-pixels **11b**, one first color sub-pixel **11a**, and one third color sub-pixel **11c** which are disposed adjacently can constitute one pixel repetition unit **10**. At the time of display, the second color sub-pixels **11b** may be obtained by rendering the brightness of sub-pixels adjacent to the second color sub-pixels **11b** and having different light-emitting colors from the second color sub-pixels **11b**. With the preceding pixel arrangement and pixel rendering technique, by borrowing the brightness of surrounding sub-pixels, the display panel can reach a higher resolution with a smaller number of sub-pixels **11**.

(93) Similarly, as shown in FIG. 12, exemplarily, first sub-pixel rows **31** and second sub-pixel rows **32** are alternately arranged along the column direction. A first sub-pixel row **31** may be formed by alternately arranging first color sub-pixels **11a** and third color sub-pixels **11c** along the row direction. A second sub-pixel row **32** may be formed by alternately arranging second color sub-pixels **11b** and third color sub-pixels **11c** along the row direction. Moreover, two adjacent sub-pixels **11** along the column direction have different emitted colors. For example, in FIG. 12, first color sub-pixels **11a** and third color sub-pixels **11c** are alternately arranged along the column direction, and second color sub-pixels **11b** and third color sub-pixels **11c** are alternately arranged along the column direction.

(94) In this case, as shown in FIG. 12, a  $2 \times 2$  sub-pixel matrix formed by two third color sub-pixels **11c**, one first color sub-pixel **11a**, and one second color sub-pixel **11b** which are disposed adjacently can constitute one pixel repetition unit **10**. At the time of display, the third color sub-pixels **11c** may be obtained by rendering the brightness of sub-pixels adjacent to the third color sub-pixels **11c** and having different emitted colors from the third color sub-pixels **11c**. With the preceding pixel arrangement and pixel rendering technique, by rendering the brightness of surrounding sub-pixels, the display panel can reach a higher resolution with a smaller number of sub-pixels **11**.

(95) In this embodiment, the maximum capacitance sub-pixel in a first sub-pixel row **31** is the first color sub-pixel **11a**. By setting the delay time  $\Delta t$  corresponding to the first sub-pixel row **31** to be greater than or equal to the charging time of the pixel capacitor of the light-emitting element **112** in the first color sub-pixel **11a**, it can be ensured that the first sub-pixel row **31** completes the charging processes of pixel capacitors of light-emitting elements **112** in all sub-pixels **11** in the row in the first light-emitting sub-stage **T31**. The maximum capacitance sub-pixel in a second sub-pixel row **32** is the second color sub-pixel **11b**. By setting the delay time  $\Delta t$  corresponding to the second sub-pixel row **32** to be greater than or equal to the charging time of the pixel capacitor of the light-emitting element **112** in the second color sub-pixel **11b**, it can be ensured that the second sub-pixel row **32** completes the charging processes of pixel capacitors of light-emitting elements **112** in all sub-pixels **11** in the row in the first light-emitting sub-stage **T31**, thereby ensuring that each row of sub-pixels **11** can emit light simultaneously when a cathode switch transistor **M0** correspondingly connected to the each row of sub-pixels **11** is turned on. Thus, the light-emitting duration of light-emitting elements **112** emitting light of different colors in the same row of sub-pixels **11** is consistent in the light emission stage **T3** so that the luminescence efficiency of the light-emitting



elements **112** tends to be consistent, thereby solving the color cast problem caused by the difference in the pixel capacitors of the light-emitting elements **112** emitting light of different colors.

(96) Meanwhile, as shown in FIG. **10** and FIG. **11**, by setting the delay time  $\Delta t$  corresponding to the second sub-pixel row **32** to be less than the delay time  $\Delta t$  corresponding to the first sub-pixel row **31**, it is possible to increase the light-emitting duration of light-emitting elements **112** of sub-pixels **11** in the second sub-pixel row **32** in the light emission stage T3 while the color cast problem caused by the difference in pixel capacitors of light-emitting elements **112** emitting light of different colors is solved, thereby improving the luminescence efficiency of sub-pixels **11** in the second sub-pixel row **32**. Moreover, a smaller working current density can be adopted under the condition of implementing the same brightness, which is helpful to prolong the service life of sub-pixels **11** in the second sub-pixel row **32**.

(97) FIG. **13** is a diagram illustrating the structure of another display panel according to an embodiment of the present disclosure. FIG. **14** is a diagram illustrating the structure of another display panel according to an embodiment of the present disclosure. FIG. **15** is a diagram illustrating the structure of another display panel according to an embodiment of the present disclosure. As shown in FIGS. **13** to **15**, optionally, first sub-pixel rows **31** and second sub-pixel rows **32** are alternately arranged along the column direction. Multiple sub-pixels **11** include first color sub-pixels **11a**, second color sub-pixels **11b**, and third color sub-pixels **11c**. The capacitance value of the pixel capacitor of the light-emitting element **112** in a first color sub-pixel **11a** is greater than the capacitance value of the pixel capacitor of the light-emitting element **112** in a second color sub-pixel **11b**. The capacitance value of the pixel capacitor of the light-emitting element **112** in the second color sub-pixel **11b** is greater than the capacitance value of the pixel capacitor of the light-emitting element **112** in a third color sub-pixel **11c**. A first sub-pixel row **31** includes first color sub-pixels **11a** and second color sub-pixels **11b**. A second sub-pixel row **32** includes third color sub-pixels **11c**. The delay time  $\Delta t$  corresponding to the first sub-pixel row **31** is greater than or equal to the charging time of the pixel capacitor of the light-emitting element **112** in a first color sub-pixel **11a**. The delay time  $\Delta t$  corresponding to the second sub-pixel row **32** is greater than or equal to the charging time of the pixel capacitor of the light-emitting element **112** in a third color sub-pixel **11c**. Alternatively, a first sub-pixel row **31** includes first color sub-pixels **11a** and third color sub-pixels **11c**. A second sub-pixel row **32** includes second color sub-pixels **11b**. The delay time  $\Delta t$  corresponding to the first sub-pixel row **31** is greater than or equal to the charging time of the pixel capacitor of the light-emitting element **112** in a first color sub-pixel **11a**. The delay time  $\Delta t$  corresponding to the second sub-pixel row **32** is greater than or equal to the charging time of the pixel capacitor of the light-emitting element **112** in a second color sub-pixel **11b**. Alternatively, a first sub-pixel row **31** includes first color sub-pixels **11a**. A second sub-pixel row **32** includes second color sub-pixels **11b** and third color sub-pixels **11c**. The delay time  $\Delta t$  corresponding to the first sub-pixel row **31** is greater than or equal to the charging time of the pixel capacitor of the light-emitting element **112** in a first color sub-pixel **11a**. The delay time  $\Delta t$  corresponding to the second sub-pixel row **32** is greater than or equal to the charging time of the pixel capacitor of the light-emitting element **112** in a second color sub-pixel **11b**.

(98) Exemplarily, as shown in FIG. **13**, first sub-pixel rows **31** and second sub-pixel rows **32** are alternately arranged along the column direction. A first sub-pixel row **31** may be formed by alternately arranging first color sub-pixels **11a** and second color sub-pixels **11b** along the row direction. A second sub-pixel row **32** may be formed by multiple third color sub-pixels **11c** arranged along the row direction. A first color sub-pixel **11a**, a second color sub-pixel **11b**, and a third color sub-pixel **11c** which are disposed adjacently can constitute one pixel repetition unit **10**. In one pixel repetition unit **10**, along the column direction, the third color sub-pixel **11c** located in a second sub-pixel row **32** may at least separately partially overlap the first color sub-pixel **11a** and the second color sub-pixel **11b** which are located in a first sub-pixel row **31**. In this arrangement manner, the arrangement of three pixels in the same pixel repetition unit **10** is more compact.

Compared with the arrangement in which the first color sub-pixel **11a**, the second color sub-pixel **11b**, and the third color sub-pixel **11c** are disposed in the same row in FIG. 1, this arrangement facilitates the improvement of the aperture ratio and screen-to-body ratio of the display panel, thereby facilitating the improvement of the display quality.

(99) In this embodiment, as shown in FIG. 10, FIG. 11, and FIG. 13, the maximum capacitance sub-pixel in a first sub-pixel row **31** is the first color sub-pixel **11a**. By setting the delay time  $\Delta t$  corresponding to the first sub-pixel row **31** to be greater than or equal to the charging time of the pixel capacitor of the light-emitting element **112** in the first color sub-pixel **11a**, it can be ensured that the first sub-pixel row **31** completes the charging processes of pixel capacitors of light-emitting elements **112** in all sub-pixels **11** in the row in the first light-emitting sub-stage T31. The maximum capacitance sub-pixel in a second sub-pixel row **32** is the third color sub-pixel **11c**. By setting the delay time  $\Delta t$  corresponding to the second sub-pixel row **32** to be greater than or equal to the charging time of the pixel capacitor of the light-emitting element **112** in the third color sub-pixel **11c**, it can be ensured that the second sub-pixel row **32** completes the charging processes of pixel capacitors of light-emitting elements **112** in all sub-pixels **11** in the row in the first light-emitting sub-stage T31, thereby ensuring that each row of sub-pixels **11** can emit light simultaneously when a cathode switch transistor M0 correspondingly connected to the each row of sub-pixels **11** is turned on. Thus, the light-emitting duration of light-emitting elements **112** emitting light of different colors in the same row of sub-pixels **11** is consistent in the light emission stage T3 so that the luminescence efficiency of the light-emitting elements **112** tends to be consistent, thereby solving the color cast problem caused by the difference in the pixel capacitors of the light-emitting elements **112** emitting light of different colors.

(100) In another embodiment, as shown in FIG. 14, first sub-pixel rows **31** and second sub-pixel rows **32** are alternately arranged along the column direction. A first sub-pixel row **31** may be formed by alternately arranging first color sub-pixels **11a** and third color sub-pixels **11c** along the row direction. A second sub-pixel row **32** may be formed by multiple second color sub-pixels **11b** arranged along the row direction. A first color sub-pixel **11a**, a second color sub-pixel **11b**, and a third color sub-pixel **11c** which are disposed adjacently can constitute one pixel repetition unit **10**. In one pixel repetition unit **10**, along the column direction, the second color sub-pixel **11b** located in a second sub-pixel row **32** may at least separately partially overlap the first color sub-pixel **11a** and the third color sub-pixel **11c** which are located in a first sub-pixel row **31**. In this arrangement manner, the arrangement of three pixels in the same pixel repetition unit **10** is more compact. Compared with the arrangement in which the first color sub-pixel **11a**, the second color sub-pixel **11b**, and the third color sub-pixel **11c** are disposed in the same row in FIG. 1, this arrangement facilitates the improvement of the aperture ratio and screen-to-body ratio of the display panel, thereby facilitating the improvement of the display quality.

(101) In this embodiment, as shown in FIG. 10, FIG. 11, and FIG. 14, the maximum capacitance sub-pixel in a first sub-pixel row **31** is the first color sub-pixel **11a**. By setting the delay time  $\Delta t$  corresponding to the first sub-pixel row **31** to be greater than or equal to the charging time of the pixel capacitor of the light-emitting element **112** in the first color sub-pixel **11a**, it can be ensured that the first sub-pixel row **31** completes the charging processes of pixel capacitors of light-emitting elements **112** in all sub-pixels **11** in the row in the first light-emitting sub-stage T31. The maximum capacitance sub-pixel in a second sub-pixel row **32** is the second color sub-pixel **11b**. By setting the delay time  $\Delta t$  corresponding to the second sub-pixel row **32** to be greater than or equal to the charging time of the pixel capacitor of the light-emitting element **112** in the second color sub-pixel **11b**, it can be ensured that the second sub-pixel row **32** completes the charging processes of pixel capacitors of light-emitting elements **112** in all sub-pixels **11** in the row in the first light-emitting sub-stage T31, thereby ensuring that each row of sub-pixels **11** can emit light simultaneously when a cathode switch transistor M0 correspondingly connected to the each row of sub-pixels **11** is turned on. Thus, the light-emitting duration of light-emitting elements **112** emitting light of

different colors in the same row of sub-pixels **11** is consistent in the light emission stage T3 so that the luminescence efficiency of the light-emitting elements **112** tends to be consistent, thereby solving the color cast problem caused by the difference in the pixel capacitors of the light-emitting elements **112** emitting light of different colors.

(102) In another embodiment, as shown in FIG. 15, first sub-pixel rows **31** and second sub-pixel rows **32** are alternately arranged along the column direction. A first sub-pixel row **31** may be formed by multiple first color sub-pixels **11a** arranged along the row direction. A second sub-pixel row **32** may be formed by alternately arranging second color sub-pixels **11b** and third color sub-pixels **11c** along the row direction. A first color sub-pixel **11a**, a second color sub-pixel **11b**, and a third color sub-pixel **11c** which are disposed adjacently can constitute one pixel repetition unit **10**. In one pixel repetition unit **10**, along the column direction, the first color sub-pixel **11a** located in a first sub-pixel row **31** may at least separately partially overlap the second color sub-pixel **11b** and the third color sub-pixel **11c** which are located in a second sub-pixel row **32**. In this arrangement manner, the arrangement of three pixels in the same pixel repetition unit **10** is more compact. Compared with the arrangement in which the first color sub-pixel **11a**, the second color sub-pixel **11b**, and the third color sub-pixel **11c** are disposed in the same row in FIG. 1, this arrangement facilitates the improvement of the aperture ratio and screen-to-body ratio of the display panel, thereby facilitating the improvement of the display quality.

(103) In this embodiment, as shown in FIG. 10, FIG. 11, and FIG. 15, the maximum capacitance sub-pixel in a first sub-pixel row **31** is the first color sub-pixel **11a**. By setting the delay time  $\Delta t$  corresponding to the first sub-pixel row **31** to be greater than or equal to the charging time of the pixel capacitor of the light-emitting element **112** in the first color sub-pixel **11a**, it can be ensured that the first sub-pixel row **31** completes the charging processes of pixel capacitors of light-emitting elements **112** in all sub-pixels **11** in the row in the first light-emitting sub-stage T31. The maximum capacitance sub-pixel in a second sub-pixel row **32** is the second color sub-pixel **11b**. By setting the delay time  $\Delta t$  corresponding to the second sub-pixel row **32** to be greater than or equal to the charging time of the pixel capacitor of the light-emitting element **112** in the second color sub-pixel **11b**, it can be ensured that the second sub-pixel row **32** completes the charging processes of pixel capacitors of light-emitting elements **112** in all sub-pixels **11** in the row in the first light-emitting sub-stage T31, thereby ensuring that each row of sub-pixels **11** can emit light simultaneously when a cathode switch transistor M0 correspondingly connected to the each row of sub-pixels **11** is turned on. Thus, the light-emitting duration of light-emitting elements **112** emitting light of different colors in the same row of sub-pixels **11** is consistent in the light emission stage T3 so that the luminescence efficiency of the light-emitting elements **112** tends to be consistent, thereby solving the color cast problem caused by the difference in the pixel capacitors of the light-emitting elements **112** emitting light of different colors.

(104) Meanwhile, as shown in FIG. 10 and FIG. 11, in the display panel shown in FIGS. 13 to 15, by setting the delay time  $\Delta t$  corresponding to the second sub-pixel row **32** to be less than the delay time  $\Delta t$  corresponding to the first sub-pixel row **31**, the light-emitting duration of light-emitting elements **112** of sub-pixels **11** in the second sub-pixel row **32** in the light emission stage T3 can be increased while the color cast problem caused by the difference in pixel capacitors of light-emitting elements **112** emitting light of different colors is solved, thereby improving the luminescence efficiency of sub-pixels **11** in the second sub-pixel row **32**. Moreover, a smaller working current density can be adopted under the condition of implementing the same brightness, which is helpful to prolong the service life of sub-pixels **11** in the second sub-pixel row **32**.

(105) It is to be noted that in the preceding embodiments, only the setting manners of the delay time  $\Delta t$  corresponding to each row of sub-pixels **11** in several pixel arrangement manners are illustrated by example. However, this is not limited to the preceding embodiments. It is to be understood that those skilled in the art can adjust the delay time  $\Delta t$  corresponding to each row of sub-pixels **11** according to a specific pixel arrangement manner to implement the desired result of

the technical solution of the present disclosure.

(106) With continued reference to FIGS. **1**, **9**, and **12** to **15**, optionally, the first color sub-pixel **11a** is a green sub-pixel G, the second color sub-pixel **11b** is a red sub-pixel R, and the third color sub-pixel **11c** is a blue sub-pixel B.

(107) It is further found that the capacitance value of the pixel capacitor of the light-emitting element **112** in the green sub-pixel G is greater than the capacitance value of the pixel capacitor of the light-emitting element **112** in the red sub-pixel R, and the capacitance value of the pixel capacitor of the light-emitting element **112** in the red sub-pixel R is greater than the capacitance value of the pixel capacitor of light-emitting element **112** in the blue sub-pixel B. Thus, in the light emission stage T3, the time required for charging the pixel capacitor of the light-emitting element **112** in the blue sub-pixel B is shorter, while the time required for charging the pixel capacitor of the light-emitting element **112** in the green sub-pixel G is longer, so that in the limited duration of the light emission stage T3 within a frame, the light-emitting duration of the light-emitting element **112** in the blue sub-pixel B is longer, and the light-emitting duration of the light-emitting element **112** in the green sub-pixel G is shorter. This results that the luminescence efficiency of the blue sub-pixel B is higher under low grayscale, the luminescence efficiency of the green sub-pixel G is lower under low grayscale, and the image is bluish.

(108) Therefore, in any one of the preceding embodiments, by setting the green sub-pixel G as the first color sub-pixel **11a**, the red sub-pixel R as the second color sub-pixel **11b**, and the blue sub-pixel B as the third color sub-pixel **11c**, the color cast problem caused by sequential reduction of the capacitance values of the pixel capacitors of the light-emitting elements **112** in the green sub-pixel G, red sub-pixel R, and blue sub-pixel B can be solved.

(109) It is to be noted that the capacitance value of the pixel capacitor of the light-emitting element **112** depends on the material of the light-emitting layer **22** in the light-emitting element **112**.

Therefore, when materials of light-emitting layers **22** in the light-emitting elements **112** in the green sub-pixel G, red sub-pixel R, and blue sub-pixel B use other material systems, the capacitance values of the pixel capacitors of the light-emitting elements **112** in the green sub-pixel G, red sub-pixel R, and blue sub-pixel B may present a different magnitude relationship from this embodiment. Thus, the green sub-pixel G, the red sub-pixel R, and the blue sub-pixel B have a different correspondence relationship with the first color sub-pixel **11a**, the second color sub-pixel **11b**, and the third color sub-pixel **11c** in the preceding embodiments. The details are not repeated here.

(110) FIG. **16** is a diagram illustrating the structure of another display panel according to an embodiment of the present disclosure. As shown in FIG. **16**, optionally, the display panel provided in this embodiment of the present disclosure includes a display region **40** and a non-display region **41** located on at least one side of the display region **40**. The non-display region **41** includes a first scan driving circuit **42** and a second scan driving circuit **43**. First light emission control signal lines EM1 are electrically connected to the first scan driving circuit **42**. Second light emission control signal lines EM2 are electrically connected to the second scan driving circuit **43**.

(111) Specifically, as shown in FIG. **16**, the first scan driving circuit **42** and the second scan driving circuit **43** are disposed in the non-display region **41** to prevent the first scan driving circuit **42** and the second scan driving circuit **43** from influencing the image display of the display region **40**. The first scan driving circuit **42** is configured to sequentially provide a first light emission control signal Emit1 for a first light emission control signal line EM1 in each row, thereby controlling light emission control transistors M16 of pixel driving circuits **111** in each row of sub-pixels **11** to be turned on. The second scan driving circuit **43** is configured to sequentially provide a second light emission control signal Emit2 for a second light emission control signal line EM2 in each row, thereby controlling cathode switch transistors M0 of pixel driving circuits **111** in each row of sub-pixels **11** to be turned on.

(112) In this embodiment, the first light emission control signal Emit1 of the first light emission

control signal line EM1 and the second light emission control signal Emit 2 of the second light emission control signal line EM2 are provided through two different scan driving circuits (for example, the first scan driving circuit 42 and the second scan driving circuit 43 in FIG. 16), respectively. The turn-on time of the light emission control transistor M16 of a pixel driving circuit 111 in each row and the turn-on time of the cathode switch transistor M0 of the pixel driving circuit 111 in each row can be independently controlled through two different scan driving circuits in a one-to-one manner. Thus, the delay time  $\Delta t$  corresponding to each row of sub-pixels 11 can be flexibly adjusted according to specific capacitance values of pixel capacitors of light-emitting elements 112 in sub-pixels 11 emitting light of different colors so that the delay time  $\Delta t$  corresponding to each row of sub-pixels 11 has a large settable range, thereby being applicable to light-emitting elements 112 of various different material systems.

(113) With continued reference to FIG. 16, optionally, the first scan driving circuit 42 and the second scan driving circuit 43 may be located on different sides of the display region 40. Thus, the width of the bezel at a single side can be reduced, thereby facilitating the narrow bezel design of the display panel.

(114) FIG. 17 is a diagram illustrating the structure of another display panel according to an embodiment of the present disclosure. As shown in FIG. 17, optionally, a first scan driving circuit 42 and a second scan driving circuit 43 may be located on the same side of a display region 40. In this case, a first light emission control signal Emit1 provided by the first scan driving circuit 42 and a second light emission control signal Emit2 provided by the second scan driving circuit 43 are transmitted from the same side of the display region 40 to the light emission control transistor M16 and the cathode switch transistor M0 of a pixel driving circuit 111 at each position in the same row. Thus, the time difference (that is, the delay time  $\Delta t$  or the duration of a first light-emitting sub-stage T31) between receiving the first light emission control signal Emit1 and the second light emission control signal Emit2 by a pixel driving circuit 111 at each position in the same row can be kept consistent, thereby ensuring that the light-emitting duration of sub-pixels 11 in the same row is consistent in the light emission stage T3, and the luminescence efficiency of light-emitting elements 112 is more consistent to further improve the color cast problem.

(115) FIG. 18 is a diagram illustrating the structure of another display panel according to an embodiment of the present disclosure. As shown in FIG. 18, optionally, the delay time  $\Delta t$  corresponding to each row of sub-pixels 11 is equal. The display panel includes N rows of sub-pixels 11. A first scan driving circuit 42 includes N cascaded first shift registers 421. A second scan driving circuit 43 includes N cascaded second shift registers 431. A first light emission control signal line EM1 corresponding to sub-pixels 11 in the i-th row is electrically connected to a first shift register 421 in the i-th stage. A second light emission control signal line EM2 corresponding to sub-pixels 11 in the i-th row is electrically connected to a second shift register 431 in the i-th stage.  $1 \leq i \leq N$ , and i is a positive integer.

(116) Exemplarily, as shown in FIG. 18,  $N=6$  is taken as an example for illustration. The display panel includes 6 rows of sub-pixels 11. The first scan driving circuit 42 includes 6 cascaded first shift registers 421. The 6 first shift registers 421 are electrically connected to 6 first light emission control signal lines EM1 correspondingly connected to the 6 rows of sub-pixels 11 in a one-to-one manner. During one frame, a first shift register 421 in the 1st stage outputs a first light emission control signal Emit1 to the 1st first light emission control signal line EM1. Thus, light emission control transistors M16 of pixel driving circuits 111 in the sub-pixels 11 in the 1st row are turned on. Meanwhile, the first light emission control signal Emit1 output from the first shift register 421 in the 1st stage also serves as a shift control signal of a first shift register 421 in the 2nd stage. That is, the first shift register 421 in the 1st stage outputs the shift control signal to the first shift register 421 in the 2nd stage while outputting the first light emission control signal Emit1 to the 1st first light emission control signal line EM1. Then, under the drive of the shift control signal, the first shift register 421 in the 2nd stage outputs the first light emission control signal Emit1 to the 2nd

first light emission control signal line EM1. Thus, light emission control transistors M16 of pixel driving circuits 111 in the sub-pixels 11 in the 2nd row are turned on. Meanwhile, the shift control signal is output to a first shift register 421 in the 3rd stage. By analogy, under the drive of the shift control signal, a first shift register 421 in the N-th stage outputs the first light emission control signal Emit1 to the N-th first light emission control signal line EM1. Thus, light emission control transistors M16 of pixel driving circuits 111 in the sub-pixels 11 in the N-th row are turned on, thereby implementing the row-by-row turn-on of light emission control transistors M16 in the N rows of sub-pixels 11.

(117) Similarly, with continued reference to FIG. 18, the second scan driving circuit 43 includes 6 cascaded second shift registers 431. The 6 second shift registers 431 are electrically connected to 6 second light emission control signal lines EM2 correspondingly connected to the 6 rows of sub-pixels 11 in a one-to-one manner. During one frame, a second shift register 431 in the 1st stage outputs a second light emission control signal Emit2 to the 1st second light emission control signal line EM2. Thus, a cathode switch transistor M0 correspondingly connected to the sub-pixels 11 in the 1st row is turned on. Meanwhile, the second light emission control signal Emit2 output from the second shift register 431 in the 1st stage also serves as a shift control signal of a second shift register 431 in the 2nd stage. That is, the second shift register 431 in the 1st stage outputs the shift control signal to the second shift register 431 in the 2nd stage while outputting the second light emission control signal Emit2 to the 1st second light emission control signal line EM2. Then, under the drive of the shift control signal, the second shift register 431 in the 2nd stage outputs the second light emission control signal Emit2 to the 2nd second light emission control signal line EM2. Thus, a cathode switch transistor M0 correspondingly connected to the sub-pixels 11 in the 2nd row is turned on. Meanwhile, the shift control signal is output to a second shift register 431 in the 3rd stage. By analogy, under the drive of the shift control signal, a second shift register 431 in the N-th stage outputs the second light emission control signal Emit2 to the N-th second light emission control signal line EM2. Thus, a cathode switch transistor M0 correspondingly connected to the sub-pixels 11 in the N-th row is turned on, thereby implementing the row-by-row turn-on of cathode switch transistors M0 correspondingly connected to N rows of sub-pixels 11.

(118) The turn-on time of the light emission control transistor M16 of a pixel driving circuit 111 in each row and the turn-on time of the cathode switch transistor M0 of the pixel driving circuit 111 in each row can be independently controlled through the first scan driving circuit 42 and the second scan driving circuit 43, respectively. Thus, the delay time  $\Delta t$  corresponding to each row of sub-pixels 11 can be flexibly adjusted according to specific capacitance values of pixel capacitors of light-emitting elements 112 in sub-pixels 11 emitting light of different colors so that the delay time  $\Delta t$  corresponding to each row of sub-pixels 11 has a large settable range, thereby being applicable to light-emitting elements 112 of various different material systems.

(119) It is to be noted that when the delay time  $\Delta t$  corresponding to each row of sub-pixels 11 is equal, only one set of N cascaded first shift registers 421 is required to implement the output of the first light emission control signal Emit1. Similarly, only one set of N cascaded second shift registers 431 is required to implement the output of the second light emission control signal Emit2. In this manner, the progressive turn-on of light emission control transistors M16 in the N rows of sub-pixels 11 and the row-by-row turn-on of cathode switch transistors M0 correspondingly connected to the N rows of sub-pixels 11 can be implemented through fewer shift registers, thereby reducing the width of the bezel and facilitating the narrow bezel design of the display panel.

(120) In addition, in practical application, the display panel has more rows of sub-pixels 11, that is, N may be a larger value. The value of N is related to the size and resolution of the display panel and may be set according to actual requirements. The details are not repeated in subsequent embodiments.

(121) FIG. 19 is a diagram illustrating the structure of another display panel according to an embodiment of the present disclosure. As shown in FIG. 19, optionally, the delay time  $\Delta t$

corresponding to each row of sub-pixels **11** is equal. The display panel includes  $N$  rows of sub-pixels **11**. A first scan driving circuit **42** and a second scan driving circuit **43** are the same scan driving circuit. The scan driving circuit includes  $N$  cascaded third shift registers **400**. A first light emission control signal line EM1 corresponding to sub-pixels **11** in the  $i$ -th row is electrically connected to a third shift register **400** in the  $i$ -th stage. A first light emission control signal line EM1 corresponding to sub-pixels **11** in the  $N$ -th row is electrically connected to a third shift register **400** in the  $N$ -th stage. A second light emission control signal line EM2 corresponding to sub-pixels **11** in the  $i$ -th row is electrically connected to a third shift register **400** in the  $(i+1)$ -th stage. A second light emission control signal line EM2 corresponding to sub-pixels **11** in the  $N$ -th row is electrically connected to a third shift register **400** in the 1st stage.  $1 \leq i \leq N-1$ , and  $i$  is a positive integer.

(122) Exemplarily, as shown in FIG. **19**,  $N=6$  is taken as an example for description. The display panel includes 6 rows of sub-pixels **11**. The first scan driving circuit **42** and the second scan driving circuit **43** are the same scan driving circuit. That is, the first scan driving circuit **42** and the second scan driving circuit **43** share the same set of 6 cascaded third shift registers **400**. The 6 third shift registers **400** are electrically connected to 6 first light emission control signal lines EM1 correspondingly connected to 6 rows of sub-pixels **11** in a one-to-one manner. The 6 third shift registers **400** are electrically connected to 6 second light emission control signal lines EM2 correspondingly connected to 6 rows of sub-pixels **11** in a one-to-one manner. A first light emission control signal line EM1 corresponding to each row of sub-pixels **11** is electrically connected to the third shift register **400** in the stage corresponding to the each row of sub-pixels **11**. A second light emission control signal line EM2 corresponding to sub-pixels **11** in each of the 1st row to the  $(N-1)$ -th row is electrically connected to a third shift register **400** in the next stage. A second light emission control signal line EM2 corresponding to sub-pixels **11** in the  $N$ -th row is electrically connected to a third shift register **400** in the 1st stage.

(123) During one frame, the third shift register **400** in the 1st stage outputs a first light emission control signal Emit1 to the 1st first light emission control signal line EM1. Thus, light emission control transistors M16 of pixel driving circuits **111** in the sub-pixels **11** in the 1st row are turned on. Then, under the drive of the first light emission control signal Emit1 (that is, a shift control signal) output from the third shift register **400** in the 1st stage, a third shift register **400** in the 2nd stage outputs the first light emission control signal Emit1 to the 2nd first light emission control signal line EM1. Thus, light emission control transistors M16 of pixel driving circuits **111** in sub-pixels **11** in the 2nd row are turned on. Meanwhile, the first light emission control signal Emit1 output from the third shift register **400** in the 2nd stage also serves as a second light emission control signal Emit2 transmitted by the 1st second light emission control signal line EM2. That is, the third shift register **400** in the 2nd stage outputs the second light emission control signal Emit2 to the 1st second light emission control signal line EM2 while outputting the first light emission control signal Emit1 to the 2nd first light emission control signal line EM1. Thus, the cathode switch transistor M0 correspondingly connected to the sub-pixels **11** in the 1st row is turned on. By analogy, under the drive of a first light emission control signal Emit1 (that is, a shift control signal) output from a third shift register **400** in the  $(N-1)$ -th stage, the third shift register **400** in the  $N$ -th stage outputs the first light emission control signal Emit1 to the  $N$ -th first light emission control signal line EM1. Thus, light emission control transistors M16 of pixel driving circuits **111** in sub-pixels **11** in the  $N$ -th row are turned on. Meanwhile, the second light emission control signal Emit2 is output to the  $(N-1)$ -th second light emission control signal line EM2 so that the cathode switch transistor M0 correspondingly connected to sub-pixels **11** in the  $(N-1)$ -th row is turned on.

(124) It is to be noted that the second light emission control signal line EM2 corresponding to the sub-pixels **11** in the  $N$ -th row is electrically connected to the third shift register **400** in the 1st stage. That is, the third shift register **400** in the 1st stage outputs the first light emission control signal Emit1 to the 1st first light emission control signal line EM1. Thus, the second light emission

control signal Emit2 is output to the N-th second light emission control signal line EM2 while light emission control transistors M16 of pixel driving circuits 111 in the sub-pixels 11 in the 1st row are turned on. Thus, the cathode switch transistor M0 correspondingly connected to the sub-pixels 11 in the N-th row is turned on. In this manner, the row-by-row turn-on of light emission control transistors M16 in the N rows of sub-pixels 11 and the row-by-row turn-on of cathode switch transistors M0 correspondingly connected to the N rows of sub-pixels 11 can be implemented.

(125) In this embodiment, when the delay time  $\Delta t$  corresponding to each row of sub-pixels 11 is equal, only one set of N cascaded third shift registers 400 is required to implement the output of the first light emission control signal Emit1 and output of the second light emission control signal Emit2. In this manner, the row-by-row turn-on of light emission control transistors M16 in the N rows of sub-pixels 11 and the row-by-row turn-on of cathode switch transistors M0 correspondingly connected to the N rows of sub-pixels 11 can be implemented through fewer shift registers, thereby further reducing the width of the bezel and facilitating the narrow bezel design of the display panel.

(126) FIG. 20 is a diagram illustrating the structure of another display panel according to an embodiment of the present disclosure. As shown in FIG. 20, optionally, multiple sub-pixels 11 include first sub-pixel rows 31 and second sub-pixel rows 32. The maximum capacitance sub-pixel in a first sub-pixel row 31 and the maximum capacitance sub-pixel in a second sub-pixel row 32 have different emitted colors. The capacitance value of the pixel capacitor of the maximum capacitance sub-pixel in the first sub-pixel row 31 is greater than the capacitance value of the pixel capacitor of the maximum capacitance sub-pixel in the second sub-pixel row 32. The delay time  $\Delta t$  corresponding to the first sub-pixel row 31 is greater than the delay time  $\Delta t$  corresponding to the second sub-pixel row 32. A second light emission control signal line EM2 includes a first light emission control signal sub-line EM21 and a second light emission control signal sub-line EM22. The gate of the cathode switch transistor M0 corresponding to the first sub-pixel row 31 is electrically connected to the first light emission control signal sub-line EM21. The gate of the cathode switch transistor M0 corresponding to the second sub-pixel row 32 is electrically connected to the second light emission control signal sub-line EM22. A second scan driving circuit 43 includes a first scan sub-circuit 51 and a second scan sub-circuit 52. The first scan sub-circuit 51 is electrically connected to the first light emission control signal sub-line EM21. The second scan sub-circuit 52 is electrically connected to the second light emission control signal sub-line EM22.

(127) In one row of sub-pixels 11, a sub-pixel 11 with the maximum capacitance value of the pixel capacitor of a light-emitting element 112 is the maximum capacitance sub-pixel in the one row of sub-pixels 11.

(128) As described above, when the maximum capacitance sub-pixel in the first sub-pixel row 31 and the maximum capacitance sub-pixel in the second sub-pixel row 32 have emitted colors, and the capacitance value of the pixel capacitor of the maximum capacitance sub-pixel in the first sub-pixel row 31 is greater than the capacitance value of the pixel capacitor of the maximum capacitance sub-pixel in the second sub-pixel row 32, the delay time  $\Delta t$  corresponding to the first sub-pixel row 31 may be set to be greater than the delay time  $\Delta t$  corresponding to the second sub-pixel row 32. Thus, the light-emitting duration of light-emitting elements 112 of sub-pixels 11 in the second sub-pixel row 32 in a light emission stage T3 can be greater than the light-emitting duration of light-emitting elements 112 of sub-pixels 11 in the first sub-pixel row 31 in the light emission stage T3 while it is ensured that the first sub-pixel row 31 completes the charging processes of pixel capacitors of light-emitting elements 112 in the first sub-pixel row 31 in a first light-emitting sub-stage T31, thereby improving the luminescence efficiency of sub-pixels 11 in the second sub-pixel row 32. Moreover, a smaller working current density can be adopted under the condition of implementing the same brightness, which is helpful to prolong the service life of sub-pixels 11 in the second sub-pixel row 32.

(129) With continued reference to FIG. 20, when the delay time  $\Delta t$  corresponding to the first sub-



pixel row **31** is different from the delay time  $\Delta t$  corresponding to the second sub-pixel row **32**, the second scan driving circuit **43** is set to include two scan sub-circuits, a first scan sub-circuit **51** and a second scan sub-circuit **52**, respectively. The first scan sub-circuit **51** is electrically connected to the gate of the cathode switch transistor **M0** corresponding to the first sub-pixel row **31** by the first light emission control signal sub-line **EM21**, so that the first scan sub-circuit **51** sequentially provides a second light emission control signal **Emit2** for first sub-pixel rows **31**, thereby controlling cathode switch transistors **M0** corresponding to the first sub-pixel rows **31** to be turned on. The second scan sub-circuit **52** is electrically connected to the gate of the cathode switch transistor **M0** corresponding to the second sub-pixel row **32** by the second light emission control signal sub-line **EM22**, so that the second scan sub-circuit **52** sequentially provides a second light emission control signal **Emit2** for second sub-pixel rows **32**, thereby controlling cathode switch transistors **M0** corresponding to the second sub-pixel rows **32** to be turned on.

(130) The second light emission control signal **Emit2** received by the cathode switch transistor **M0** corresponding to the first sub-pixel row **31** and the second light emission control signal **Emit2** received by the cathode switch transistor **M0** corresponding to the second sub-pixel row **32** are provided by two independent scan sub-circuits (for example, the first scan sub-circuit **51** and the second scan sub-circuit **52** in FIG. 20) in a one-to-one manner. The turn-on time of the cathode switch transistor **M0** corresponding to the first sub-pixel row **31** and the turn-on time of the cathode switch transistor **M0** corresponding to the second sub-pixel row **32** can be independently controlled through two different scan sub-circuits in a one-to-one manner. Thus, the delay time  $\Delta t$  corresponding to the first sub-pixel row **31** and the delay time  $\Delta t$  corresponding to the second sub-pixel row **32** can be flexibly adjusted according to different pixel arrangements so that the delay time  $\Delta t$  of the first sub-pixel row **31** and the delay time  $\Delta t$  of the second sub-pixel row **32** are different.

(131) With continued reference to FIG. 20, exemplarily, the first scan sub-circuit **51** may include multiple cascaded fourth shift registers **511**. The number of the fourth shift registers **511** is the same as the number of the first sub-pixel rows **31**. The multiple fourth shift registers **511** are electrically connected to multiple first light emission control signal sub-lines **EM21** correspondingly connected to multiple rows of first sub-pixels **31** in a one-to-one manner. The second scan sub-circuit **52** may include multiple cascaded fifth shift registers **521**. The number of the fifth shift registers **521** is the same as the number of the second sub-pixel rows **32**. The multiple fifth shift registers **521** are electrically connected to multiple second light emission control signal sub-lines **EM22** correspondingly connected to multiple rows of second sub-pixels **32** in a one-to-one manner.

(132) As shown in FIG. 20, that first sub-pixel rows **31** and second sub-pixel rows **32** are alternately arranged along the column direction is taken as an example for illustration. During one frame, a fourth shift register **511** in the 1st stage outputs a second light emission control signal **Emit2** to the 1st first light emission control signal sub-line **EM21**. Thus, a cathode switch transistor **M0** correspondingly connected to a first sub-pixel row **31** in the 1st row is turned on.

(133) Then, a fifth shift register **521** in the 1st stage outputs a second light emission control signal **Emit2** to the 1st second light emission control signal sub-line **EM22**. Thus, a cathode switch transistor **M0** correspondingly connected to a second sub-pixel row **32** in the 1st row is turned on.

(134) Then, under the drive of the second light emission control signal **Emit2** (that is, a shift control signal) output from the fourth shift register **511** in the 1st stage, a fourth shift register **511** in the 2nd stage outputs the second light emission control signal **Emit2** to a second first light emission control signal sub-line **EM21**. Thus, a cathode switch transistor **M0** correspondingly connected to a first sub-pixel row **31** in the 2nd row is turned on.

(135) Then, under the drive of the second light emission control signal **Emit2** (that is, a shift control signal) output from the fifth shift register **521** in the 1st stage, a fifth shift register **521** in the 2nd stage outputs the second light emission control signal **Emit2** to the 2nd second light emission control signal sub-line **EM22**. Thus, a cathode switch transistor **M0** correspondingly

connected to a second sub-pixel row **32** in the 2nd row is turned on.

(136) By analogy, the row-by-row turn-on of cathode switch transistors **M0** correspondingly connected to first sub-pixel rows **31** and second sub-pixel rows **32** is implemented.

(137) FIG. **21** is a diagram illustrating the structure of another display panel according to an embodiment of the present disclosure. FIG. **22** is a sectional structure diagram of FIG. **21** taken along direction C-C'. As shown in FIG. **21** and FIG. **22**, optionally, cathodes **23** of light-emitting elements **112** in the same row of sub-pixels **11** are connected in series to form a cathode row **60**. Cathode rows **60** in different rows of sub-pixels **11** are insulated from each other. The cathode row **60** is electrically connected to a second power signal line PVEE by a cathode switch transistor **M0** corresponding to the cathode row **60**.

(138) As described above, driving processes of pixel driving circuits **111** of the same row of sub-pixels **11** are performed simultaneously. That is, initialization stages **T1**, data signal voltage write stages **T2**, and light emission stages **T3** (including first light-emitting sub-stages **T31** and second light-emitting sub-stages **T32**) of the same row of sub-pixels **11** are performed simultaneously.

(139) As shown in FIG. **21** and FIG. **22**, in this embodiment, cathodes **23** of light-emitting elements **112** of the same row of sub-pixels **11** are set to be connected in series to form a cathode row **60**. Each row of sub-pixels **11** is correspondingly provided with one cathode row **60**. A cathode row **60** covers light-emitting layers **22** of light-emitting elements **112** of all sub-pixels **11** in the row along the thickness direction of the display panel.

(140) Multiple cathode rows **60** extend along the row direction and are arranged along the column direction. A gap exists between two adjacent cathode rows **60** along the column direction. Thus, the two adjacent cathode rows **60** are insulated.

(141) With continued reference to FIG. **21** and FIG. **22**, each cathode row **60** is electrically connected to the second power signal line PVEE by a corresponding cathode switch transistor **M0**. When a cathode switch transistor **M0** corresponding to a row of sub-pixels **11** is turned on, the cathode row **60** corresponding to the row of sub-pixels **11** and the second power signal line PVEE are turned on. Since cathodes **23** of all light-emitting elements **112** in the row of sub-pixels **11** are connected in series to form the cathode row **60**, the turn-on between the cathode row **60** and the second power signal line PVEE is equivalent to the turn-on between the cathodes **23** of all light-emitting elements **112** in the row of sub-pixels **11** and the second power signal line PVEE. In this manner, at least two sub-pixels **11** of a row of sub-pixels **11** can share one cathode switch transistor **M0**, so that the turn-on and turn-off between cathodes **23** of light-emitting elements **112** in the row of sub-pixels **11** and the second power signal line PVEE can be controlled through a smaller number of cathode switch transistors **M0**, which is helpful to reduce the occupation space of cathode switch transistors **M0** and reduce the cost.

(142) With continued reference to FIG. **21** and FIG. **22**, optionally, one end of a cathode row is electrically connected to the second power signal line PVEE by one cathode switch transistor **M0**. In this manner, each row of sub-pixels **11** needs only one cathode switch transistor **M0** to control the turn-on and turn-off between cathodes **23** of light-emitting elements **112** in the each row of sub-pixels **11** and the second power signal line PVEE, which is helpful to reduce the occupation space of cathode switch transistors **M0** and reduce the cost.

(143) FIG. **23** is a diagram illustrating the structure of another display panel according to an embodiment of the present disclosure. As shown in FIG. **23**, optionally, one cathode switch transistor **M0** is connected between each of two ends of a cathode row **60** and a second power signal line PVEE, respectively. In this manner, when cathode switch transistors **M0** are turned on, the turn-on between each of the two ends of the cathode row **60** and the second power signal line PVEE is achieved. Thus, the transmission efficiency of a second power voltage transmitted from the second power signal line PVEE to the cathode row **60** can be improved so that the second power voltage on the second power signal line PVEE can be rapidly transmitted to a light-emitting element **112** at each position in the same row of sub-pixels **11**, thereby helping to make the light-

emitting start time of light-emitting elements **112** in the same row of sub-pixels **11** more consistent in a light emission stage T3.

(144) With continued reference to FIG. **21** and FIG. **23**, optionally, the shortest distance between adjacent cathode rows **60** is  $d$ , where  $d \geq 2.5 \mu\text{m}$ . By setting a gap greater than or equal to  $2.5 \mu\text{m}$  between adjacent cathode rows **60**, the preparation process is easily implemented while the insulation between adjacent cathode rows **60** is ensured.

(145) With continued reference to FIG. **21** and FIG. **23**, optionally, cathode switch transistors **M0** are disposed in a non-display region **41** to prevent the cathode switch transistors **M0** from affecting the image display of a display region **40**.

(146) FIG. **24** is a partial sectional structure diagram of a display panel according to an embodiment of the present disclosure. FIG. **25** is a partial sectional structure diagram of another display panel according to an embodiment of the present disclosure. As shown in FIG. **24** and FIG. **25**, optionally, a cathode switch transistor **M0** includes at least two sub-switch transistors **M01**. The at least two sub-switch transistors **M01** are connected in series, and/or the cathode switch transistor **M0** includes a double-gate transistor.

(147) As shown in FIG. **24**, a cathode switch transistor **M0** is formed by connecting at least two sub-switch transistors **M01** in series. When the cathode switch transistor **M0** is cut off under the action of the second light emission control signal Emit2 of the second light emission control signal line EM2, all of the at least two sub-switch transistors **M01** are cut off. In this manner, the leakage current of the cathode switch transistor **M0** at cutoff can be smaller, thereby avoiding the problem of emitting undesired light by a light-emitting element **112** in a non-light emission stage caused by the leakage current between a cathode **23** and the second power signal line PVEE.

(148) In other embodiments, as shown in FIG. **25**, by setting a cathode switch transistor **M0** as a double-gate transistor, the leakage current of the cathode switch transistor **M0** at cutoff can be reduced, thereby avoiding the problem of emitting undesired light by a light-emitting element **112** in a non-light emission stage caused by the leakage current between a cathode **23** and the second power signal line PVEE.

(149) With continued reference to FIG. **24** and FIG. **25**, optionally, the gate of the cathode switch transistor **M0** is located on the same film layer as the second light emission control signal line EM2.

(150) The gate of the cathode switch transistor **M0** needs to be electrically connected to the second light emission control signal line EM2. By disposing the gate of the cathode switch transistor **M0** and the second light emission control signal line EM2 in the same layer, the gate of the cathode switch transistor **M0** can be directly connected to the second light emission control signal line EM2 in the same film layer without performing punching, thereby helping to reduce technique difficulty.

(151) Moreover, by disposing the gate of the cathode switch transistor **M0** and the second light emission control signal line EM2 in the same layer, the arrangement of one metal layer can be reduced, thereby achieving the purpose of reducing the production cost and the thickness of the substrate. The gate of the cathode switch transistor **M0** may be made of the same material as the second light emission control signal line EM2. Thus, the gate of the cathode switch transistor **M0** and the second light emission control signal line EM2 can be prepared in the same process, thereby shortening the process time.

(152) With continued reference to FIG. **24**, optionally, the display panel further includes a base substrate 00. The cathode switch transistor **M0** and a drive transistor **M3** are located on the same side of the base substrate 00. The active layer of the cathode switch transistor **M0** and the active layer **01** of the drive transistor **M3** are located on the same film layer. The gate layer of the cathode switch transistor **M0** and the gate layer **02** of the drive transistor **M3** are located on the same film layer. The source and drain electrode layer of the cathode switch transistor **M0** and the source and drain electrode layer **03** of the drive transistor **M3** are located on the same film layer.

(153) Specifically, as shown in FIGS. **2**, **22**, **24**, and **25**, the drive transistor **M3** can include the

active layer **01**, the gate layer **02**, and the source and drain electrode layer **03** which are stacked on the base substrate **00**.

(154) As shown in FIG. **24**, in this embodiment, by disposing the cathode switch transistor **M0** and the drive transistor **M3** in the same layer, the number of film layers is reduced, thereby achieving the purpose of reducing the production cost and the thickness of the substrate. Moreover, the cathode switch transistor **M0** and the drive transistor **M3** can be prepared in the same process, thereby shortening the process time.

(155) With continued reference to FIG. **24** and FIG. **25**, optionally, the display panel also includes the base substrate **00**. The cathode switch transistor **M0** and the second power signal line **PVEE** are located on the same side of the base substrate **00**. The source and drain electrode layer of the cathode switch transistor **M0** and the second power signal line **PVEE** are located on the same film layer.

(156) The source and drain electrode layer of the cathode switch transistor **M0** needs to be electrically connected to the second power signal line **PVEE**, so that the source and drain electrode layer of the cathode switch transistor **M0** and the second power signal line **PVEE** are disposed in the same layer, and thus, the source and drain electrode layer of the cathode switch transistor **M0** can be directly connected to the second power signal line **PVEE** in the same film layer without performing punching, thereby helping to reduce technique difficulty.

(157) Moreover, by disposing the source and drain electrode layer of the cathode switch transistor **M0** and the second power signal line **PVEE** in the same layer, the arrangement of a metal layer can be reduced, thereby achieving the purpose of reducing the production cost and the thickness of the substrate. The source and drain electrode layer of the cathode switch transistor **M0** may be made of the same material as the second power signal line **PVEE**. Thus, the source and drain electrode layer of the cathode switch transistor **M0** and the second power signal line **PVEE** can be prepared in the same process, thereby shortening the process time.

(158) With continued reference to FIG. **24** and FIG. **25**, optionally, the display panel further includes the base substrate **00**. The cathode switch transistor **M0** and the light-emitting element **112** are located on the same side of the base substrate **00**. The light-emitting element **112** is located on the side of the cathode switch transistor **M0** facing away from the base substrate **00**. The cathode **23** of the light-emitting element **112** is located on the side of the anode **21** of the light-emitting element **112** facing away from the base substrate **00**. The source and drain electrode layer of the cathode switch transistor **M0** is electrically connected to the cathode **23** of the light-emitting element **112** by a bridge metal layer **70**. The bridge metal layer **70** is located between the source and drain electrode layer of the cathode switch transistor **M0** and the cathode **23** of the light-emitting element **112**.

(159) The source and drain electrode layer of the cathode switch transistor **M0** is electrically connected to the cathode **23** of the light-emitting element **112** by the bridge metal layer **70** located between the source and drain electrode layer of the cathode switch transistor **M0** and the cathode **23** of the light-emitting element **112**. Thus, a deep punching process can be avoided when the source and drain electrode layer of the cathode switch transistor **M0** is electrically connected to the cathode **23** of the light-emitting element **112**, thereby ensuring a simple electrical connection process between the source and drain electrode layer of the cathode switch transistor **M0** and the cathode **23** of the light-emitting element **112**.

(160) With continued reference to FIG. **24** and FIG. **25**, optionally, the bridge metal layer **70** and the anode **21** are located in the same film layer.

(161) By disposing the bridge metal layer **70** and the anode **21** in the same layer, the arrangement of one metal layer can be reduced, thereby achieving the purpose of reducing the production cost and the thickness of the substrate. Moreover, the bridge metal layer **70** may be made of the same material as the anode **21**. Thus, the bridge metal layer **70** and the anode **21** can be prepared in the same process, thereby shortening the process time.

(162) Based on the same inventive concept, the embodiments of the present disclosure further provide a display device. FIG. 26 is a diagram illustrating the structure of a display device according to an embodiment of the present disclosure. As shown in FIG. 26, a display device 80 includes a display panel 81 according to any embodiment of the present disclosure. Therefore, the display device 80 provided in the embodiments of the present disclosure has the technical effects of the technical solution of any one of the preceding embodiments. Explanations of structures and terms the same as or corresponding to structures and terms in the preceding embodiments are not repeated here.

(163) The display device 80 provided in the embodiments of the present disclosure may be the phone shown in FIG. 26, or may be any electronic product with display function, including but not limited to a television, a laptop, a desktop display, a tablet computer, a digital camera, a smart bracelet, a smart glass, a vehicle-mounted display, medical equipment, industrial control equipment, and a touch interactive terminal. No special limitations are made thereto in the embodiments of the present disclosure.

(164) Based on the same inventive concept, the embodiments of the present disclosure also provide a driving method of a display panel. The driving method can be applied to driving any display device provided in the preceding embodiments. Explanations of structures and terms the same as or corresponding to structures and terms in the preceding embodiments are not repeated here.

(165) The display panel includes a cathode switch transistor and multiple sub-pixels arranged in an array. A sub-pixel includes a pixel driving circuit and a light-emitting element electrically connected to the pixel driving circuit. The pixel driving circuit includes a drive transistor and at least one light emission control transistor. The drive transistor, the light emission control transistor, the light-emitting element, and the cathode switch transistor are connected in series between a first power signal line and a second power signal line. The light emission control transistor is connected in series between the anode of the light-emitting element and the first power signal line. The cathode switch transistor is connected in series between the cathode of the light-emitting element and the second power signal line. The gate of the light emission control transistor is electrically connected to a first light emission control signal line. The gate of the cathode switch transistor is electrically connected to a second light emission control signal line.

(166) For the description of structures and explanations of terms of the preceding display panel, reference can be made to the preceding embodiments. The details are not repeated here.

(167) FIG. 27 is a flowchart of a driving method of a display panel according to an embodiment of the present disclosure. As shown in FIG. 27, the driving method includes the following steps.

(168) In S110, a first light emission control signal is applied to the first light emission control signal line.

(169) In S120, a second light emission control signal is applied to the second light emission control signal line.

(170) In at least one light emission stage, the start time of an effective pulse of the second light emission control signal lags behind the start time of an effective pulse of the first light emission control signal.

(171) According to the driving method of the display panel provided in the embodiments of the present disclosure, the first light emission control signal is applied to the first light emission control signal line in the light emission stage so that light emission control transistors are first turned on, and pixel capacitors of light-emitting elements are charged. After the charging of pixel capacitors of light-emitting elements emitting light of different colors is completed, the second light emission control signal is applied to the second light emission control signal line so that the cathode switch transistor is turned on. Thus, the light-emitting elements emitting light of different colors emit light simultaneously when the cathode switch transistor is turned on so that the light-emitting duration and luminescence efficiency of the light-emitting elements emitting light of different colors tend to be consistent in the light emission stage, thereby solving the color cast problem caused by the

difference in the pixel capacitors of the light-emitting elements emitting light of different colors. (172) Optionally, at least part of sub-pixels have different emitted colors. Light-emitting elements in the sub-pixels have corresponding pixel capacitors. Capacitance values of pixel capacitors of light-emitting elements in sub-pixels of different emitted colors are different. In at least one row of sub-pixels, a sub-pixel with the maximum capacitance value of the pixel capacitor of a light-emitting element is the maximum capacitance sub-pixel. The charging time of the pixel capacitor of the light-emitting element in the maximum capacitance sub-pixel is  $t_1$ . For at least one row of sub-pixels, in the light emission stage, a duration in which the start time of the effective pulse of the second light emission control signal of the second light emission control signal line corresponding to the sub-pixels lags behind the start time of the effective pulse of the first light emission control signal of the first light emission control signal line corresponding to the sub-pixels is delay time. The delay time is  $\Delta t$ , and  $\Delta t \geq t_1$ .

(173) For the description of structures and explanations of terms of the preceding display panel, reference can be made to the preceding embodiments. The details are not repeated here.

(174) In this embodiment, for any row of sub-pixels, the delay time  $\Delta t$  is set to be greater than or equal to the charging time  $t_1$  of the pixel capacitor of the light-emitting element in the maximum capacitance sub-pixel, so that for any row of sub-pixels, the charging processes of pixel capacitors of light-emitting elements of all sub-pixels in this row can be completed before cathode switch transistor is turned on, thereby ensuring that each row of sub-pixels emits light simultaneously when the cathode switch transistor correspondingly connected to the each row of sub-pixels is turned on (that is, at the start time of the effective pulse of the second light emission control signal of the second light emission control signal line) without charging the pixel capacitors of the light-emitting elements again. Thus, the light-emitting start time of light-emitting elements emitting light of different colors in the same row of sub-pixels is consistent in the light emission stage, and the light-emitting duration of the light-emitting elements emitting light of different colors tends to be consistent in the light emission stage. Moreover, the luminescence efficiency of the light-emitting elements tends to be consistent, thereby solving the color cast problem caused by the difference in the pixel capacitors of the light-emitting elements emitting light of different colors.

(175) It is to be understood that various forms of processes shown above may be adopted with steps reordered, added, or deleted. For example, the steps described in the present disclosure may be performed in parallel, sequentially, or in different orders, as long as the desired results of the technical solutions of the present disclosure can be implemented, and no limitation is imposed herein.

(176) The preceding specific embodiments do not constitute a limitation on the protection scope of the present disclosure. It is to be understood by those skilled in the art that various modifications, combinations, sub-combinations, and substitutions may be performed according to design requirements and other factors. Any modifications, equivalent replacements, improvements, and the like within the spirit and principle of the present disclosure shall fall within the protection scope of the present disclosure.

## Claims

1. A display panel, comprising: a cathode switch transistor and a plurality of sub-pixels arranged in an array, wherein a sub-pixel of the plurality of sub-pixels comprises a pixel driving circuit and a light-emitting element electrically connected to the pixel driving circuit, wherein the pixel driving circuit comprises a drive transistor and at least one light emission control transistor; the drive transistor, the at least one light emission control transistor, the light-emitting element, and the cathode switch transistor are connected in series between a first power signal line and a second power signal line, wherein the at least one light emission control transistor is connected in series between an anode of the light-emitting element and the first power signal line, and the cathode

switch transistor is connected in series between a cathode of the light-emitting element and the second power signal line; and a gate of a light emission control transistor of the at least one light emission control transistor is electrically connected to a first light emission control signal line, a gate of the cathode switch transistor is electrically connected to a second light emission control signal line, and in at least one light emission stage, start time of an effective pulse of a second light emission control signal of the second light emission control signal line lags behind start time of an effective pulse of a first light emission control signal of the first light emission control signal line; and wherein at least part of the plurality of sub-pixels have different emitted colors, a light-emitting element of light-emitting elements in the plurality of sub-pixels has a pixel capacitor corresponding to the light-emitting element, and capacitance values of pixel capacitors of light-emitting elements in sub-pixels of different emitted colors are different; in at least one row of sub-pixels of the plurality of sub-pixels, a sub-pixel with a maximum capacitance value of a pixel capacitor of a light-emitting element is a maximum capacitance sub-pixel, and charging time of the pixel capacitor of the light-emitting element in the maximum capacitance sub-pixel is  $t_1$ ; and for at least one row of sub-pixels of the plurality of sub-pixels, in the at least one light emission stage, a duration in which start time of an effective pulse of a second light emission control signal of a second light emission control signal line corresponding to the at least one row of sub-pixels lags behind start time of an effective pulse of a first light emission control signal of a first light emission control signal line corresponding to the at least one row of sub-pixels is delay time, wherein the delay time is  $\Delta t$ , and  $\Delta t \geq t_1$ .

2. The display panel according to claim 1, wherein the plurality of sub-pixels comprise a first color sub-pixel, a second color sub-pixel, and a third color sub-pixel, wherein a capacitance value of a pixel capacitor of a light-emitting element in the first color sub-pixel is greater than a capacitance value of a pixel capacitor of a light-emitting element in the second color sub-pixel, and the capacitance value of the pixel capacitor of the light-emitting element in the second color sub-pixel is greater than a capacitance value of a pixel capacitor of a light-emitting element in the third color sub-pixel; charging time of the pixel capacitor of the light-emitting element in the first color sub-pixel is  $t_2$ , wherein  $\Delta t \geq t_2$ .

3. The display panel according to claim 2, wherein delay time corresponding to each row of sub-pixels of the plurality of sub-pixels is equal.

4. The display panel according to claim 1, wherein  $2 \mu s \leq \Delta t \leq 4 \mu s$ .

5. The display panel according to claim 1, wherein the plurality of sub-pixels comprise first sub-pixel rows and second sub-pixel rows, wherein an emitted color of a maximum capacitance sub-pixel in a first sub-pixel row of the first sub-pixel rows and an emitted color of a maximum capacitance sub-pixel in a second sub-pixel row of the second sub-pixel rows are different, and a capacitance value of a pixel capacitor of the maximum capacitance sub-pixel in the first sub-pixel row is greater than a capacitance value of a pixel capacitor of the maximum capacitance sub-pixel in the second sub-pixel row; and delay time corresponding to the first sub-pixel row is greater than delay time corresponding to the second sub-pixel row.

6. The display panel according to claim 5, wherein the first sub-pixel rows and the second sub-pixel rows are alternately arranged along a column direction; the plurality of sub-pixels comprise first color sub-pixels, second color sub-pixels, and third color sub-pixels, wherein a capacitance value of a pixel capacitor of a light-emitting element in a first color sub-pixel of the first color sub-pixels is greater than a capacitance value of a pixel capacitor of a light-emitting element in a second color sub-pixel of the second color sub-pixels, and the capacitance value of the pixel capacitor of the light-emitting element in the second color sub-pixel is greater than a capacitance value of a pixel capacitor of a light-emitting element in a third color sub-pixel of the third color sub-pixels; the first sub-pixel row comprises a first color sub-pixel and a second color sub-pixel, and the second sub-pixel row comprises a second color sub-pixel and a third color sub-pixel; or, the first sub-pixel row comprises a first color sub-pixel and a third color sub-pixel, and the second sub-pixel row

- comprises a second color sub-pixel and a third color sub-pixel; and the delay time corresponding to the first sub-pixel row is greater than or equal to charging time of a pixel capacitor of a light-emitting element in the first color sub-pixel in the first sub-pixel row, and the delay time corresponding to the second sub-pixel row is greater than or equal to charging time of a pixel capacitor of a light-emitting element in the second color sub-pixel in the second sub-pixel row.
7. The display panel according to claim 5, wherein the first sub-pixel rows and the second sub-pixel rows are alternately arranged along a column direction; the plurality of sub-pixels comprise first color sub-pixels, second color sub-pixels, and third color sub-pixels, wherein a capacitance value of a pixel capacitor of a light-emitting element in a first color sub-pixel of the first color sub-pixels is greater than a capacitance value of a pixel capacitor of a light-emitting element in a second color sub-pixel of the second color sub-pixels, and the capacitance value of the pixel capacitor of the light-emitting element in the second color sub-pixel is greater than a capacitance value of a pixel capacitor of a light-emitting element in a third color sub-pixel of the third color sub-pixels; and the first sub-pixel row comprises a first color sub-pixel and a second color sub-pixel, and the second sub-pixel row comprises a third color sub-pixel; and the delay time corresponding to the first sub-pixel row is greater than or equal to charging time of a pixel capacitor of a light-emitting element in the first color sub-pixel in the first sub-pixel row, and the delay time corresponding to the second sub-pixel row is greater than or equal to charging time of a pixel capacitor of a light-emitting element in the third color sub-pixel in the second sub-pixel row; or, the first sub-pixel row comprises a first color sub-pixel and a third color sub-pixel, and the second sub-pixel row comprises a second color sub-pixel; and the delay time corresponding to the first sub-pixel row is greater than or equal to charging time of a pixel capacitor of a light-emitting element in the first color sub-pixel in the first sub-pixel row, and the delay time corresponding to the second sub-pixel row is greater than or equal to charging time of a pixel capacitor of a light-emitting element in the second color sub-pixel in the second sub-pixel row; or, the first sub-pixel row comprises a first color sub-pixel, and the second sub-pixel row comprises a second color sub-pixel and a third color sub-pixel; and the delay time corresponding to the first sub-pixel row is greater than or equal to charging time of a pixel capacitor of a light-emitting element in the first color sub-pixel in the first sub-pixel row, and the delay time corresponding to the second sub-pixel row is greater than or equal to charging time of a pixel capacitor of a light-emitting element in the second color sub-pixel in the second sub-pixel row.
8. The display panel according to claim 2, wherein the first color sub-pixel is a green sub-pixel, the second color sub-pixel is a red sub-pixel, and the third color sub-pixel is a blue sub-pixel.
9. The display panel according to claim 6, wherein the first color sub-pixel is a green sub-pixel, the second color sub-pixel is a red sub-pixel, and the third color sub-pixel is a blue sub-pixel.
10. The display panel according to claim 1, wherein the display panel further comprises a display region and a non-display region located on at least one side of the display region, and the non-display region comprises a first scan driving circuit and a second scan driving circuit; and the first light emission control signal line is electrically connected to the first scan driving circuit, and the second light emission control signal line is electrically connected to the second scan driving circuit.
11. The display panel according to claim 2, wherein the display panel further comprises a display region and a non-display region located on at least one side of the display region, and the non-display region comprises a first scan driving circuit and a second scan driving circuit; and the first light emission control signal line is electrically connected to the first scan driving circuit, and the second light emission control signal line is electrically connected to the second scan driving circuit.
12. The display panel according to claim 10, wherein delay time corresponding to each row of sub-pixels of the plurality of sub-pixels is equal; N rows of sub-pixels are provided in the display panel; the first scan driving circuit comprises N cascaded first shift registers, and the second scan driving circuit comprises N cascaded second shift registers; a first light emission control signal line corresponding to sub-pixels in an i-th row among the N rows of sub-pixels is electrically connected



to a first shift register in an  $i$ -th stage among the  $N$  cascaded first shift registers; and a second light emission control signal line corresponding to the sub-pixels in the  $i$ -th row among the  $N$  rows of sub-pixels is electrically connected to a second shift register in the  $i$ -th stage among the  $N$  cascaded first shift registers, wherein  $1 \leq i \leq N$ , and  $i$  is a positive integer.

13. The display panel according to claim 10, wherein delay time corresponding to each row of sub-pixels of the plurality of sub-pixels is equal;  $N$  rows of sub-pixels are provided in the display panel; the first scan driving circuit and the second scan driving circuit are a same scan driving circuit, and the same scan driving circuit comprises  $N$  cascaded third shift registers; a first light emission control signal line corresponding to sub-pixels in an  $i$ -th row among the  $N$  rows of sub-pixels is electrically connected to a third shift register in an  $i$ -th stage among the  $N$  cascaded third shift registers, and a first light emission control signal line corresponding to sub-pixels in an  $N$ -th row among the  $N$  rows of sub-pixels is electrically connected to a third shift register in an  $N$ -th stage among the  $N$  cascaded third shift registers; and a second light emission control signal line corresponding to the sub-pixels in the  $i$ -th row among the  $N$  rows of sub-pixels is electrically connected to a third shift register in an  $(i+1)$ -th stage among the  $N$  cascaded third shift registers, and a second light emission control signal line corresponding to the sub-pixels in the  $N$ -th row among the  $N$  rows of sub-pixels is electrically connected to a third shift register in a first stage among the  $N$  cascaded third shift registers, wherein  $1 \leq i \leq N-1$ , and  $i$  is a positive integer.

14. The display panel according to claim 10, wherein the plurality of sub-pixels comprise first sub-pixel rows and second sub-pixel rows, wherein a maximum capacitance sub-pixel in a first sub-pixel row of the first sub-pixel rows and a maximum capacitance sub-pixel in a second sub-pixel row of the second sub-pixel rows have different emitted colors, and a capacitance value of a pixel capacitor of the maximum capacitance sub-pixel in the first sub-pixel row is greater than a capacitance value of a pixel capacitor of the maximum capacitance sub-pixel in the second sub-pixel row; delay time corresponding to the first sub-pixel row is greater than delay time corresponding to the second sub-pixel row; the second light emission control signal line comprises a first light emission control signal sub-line and a second light emission control signal sub-line, a gate of a cathode switch transistor corresponding to the first sub-pixel row is electrically connected to the first light emission control signal sub-line, and a gate of a cathode switch transistor corresponding to the second sub-pixel row is electrically connected to the second light emission control signal sub-line; and the second scan driving circuit comprises a first scan sub-circuit and a second scan sub-circuit, the first scan sub-circuit is electrically connected to the first light emission control signal sub-line, the second scan sub-circuit is electrically connected to the second light emission control signal sub-line.

15. The display panel according to claim 1, wherein cathodes of light-emitting elements in sub-pixels in a same row among the plurality of sub-pixels are connected in series to form a cathode row; cathode rows in sub-pixels in different rows among the plurality of sub-pixels are insulated; and a cathode row of the cathode rows is electrically connected to the second power signal line by a cathode switch transistor corresponding to the cathode row.

16. The display panel according to claim 1, wherein the cathode switch transistor comprises at least two sub-switch transistors, and the at least two sub-switch transistors are connected in series; and/or, the cathode switch transistor comprises a double-gate transistor.

17. A display device, comprising a display panel, wherein the display panel comprises a cathode switch transistor and a plurality of sub-pixels arranged in an array, wherein a sub-pixel of the plurality of sub-pixels comprises a pixel driving circuit and a light-emitting element electrically connected to the pixel driving circuit, wherein the pixel driving circuit comprises a drive transistor and at least one light emission control transistor; the drive transistor, the at least one light emission control transistor, the light-emitting element, and the cathode switch transistor are connected in series between a first power signal line and a second power signal line, wherein the at least one light emission control transistor is connected in series between an anode of the light-emitting

element and the first power signal line, and the cathode switch transistor is connected in series between a cathode of the light-emitting element and the second power signal line; and a gate of a light emission control transistor of the at least one light emission control transistor is electrically connected to a first light emission control signal line, a gate of the cathode switch transistor is electrically connected to a second light emission control signal line, and in at least one light emission stage, start time of an effective pulse of a second light emission control signal of the second light emission control signal line lags behind start time of an effective pulse of a first light emission control signal of the first light emission control signal line; and wherein at least part of the plurality of sub-pixels have different emitted colors, a light-emitting element of light-emitting elements in the plurality of sub-pixels has a pixel capacitor corresponding to the light-emitting element, and capacitance values of pixel capacitors of light-emitting elements in sub-pixels of different emitted colors are different; in at least one row of sub-pixels of the plurality of sub-pixels, a sub-pixel with a maximum capacitance value of a pixel capacitor of a light-emitting element is a maximum capacitance sub-pixel, and charging time of the pixel capacitor of the light-emitting element in the maximum capacitance sub-pixel is  $t_1$ ; and for at least one row of sub-pixels of the plurality of sub-pixels, in the at least one light emission stage, a duration in which start time of an effective pulse of a second light emission control signal of a second light emission control signal line corresponding to the at least one row of sub-pixels lags behind start time of an effective pulse of a first light emission control signal of a first light emission control signal line corresponding to the at least one row of sub-pixels is delay time, wherein the delay time is  $\Delta t$ , and  $\Delta t \geq t_1$ .

18. A driving method of a display panel, wherein the display panel comprises a cathode switch transistor and a plurality of sub-pixels arranged in an array; a sub-pixel of the plurality of sub-pixels comprises a pixel driving circuit and a light-emitting element electrically connected to the pixel driving circuit, wherein the pixel driving circuit comprises a drive transistor and at least one light emission control transistor; the drive transistor, the at least one light emission control transistor, the light-emitting element, and the cathode switch transistor are connected in series between a first power signal line and a second power signal line, wherein the at least one light emission control transistor is connected in series between an anode of the light-emitting element and the first power signal line, and the cathode switch transistor is connected in series between a cathode of the light-emitting element and the second power signal line; and a gate of a light emission control transistor of the at least one light emission control transistor is electrically connected to a first light emission control signal line, and a gate of the cathode switch transistor is electrically connected to a second light emission control signal line; wherein at least part of the plurality of sub-pixels have different emitted colors, a light-emitting element of light-emitting elements in the plurality of sub-pixels has a pixel capacitor corresponding to the light-emitting element, and capacitance values of pixel capacitors of light-emitting elements in sub-pixels of different emitted colors are different; in at least one row of sub-pixels of the plurality of sub-pixels, a sub-pixel with a maximum capacitance value of a pixel capacitor of a light-emitting element is a maximum capacitance sub-pixel, and charging time of the pixel capacitor of the light-emitting element in the maximum capacitance sub-pixel is  $t_1$ ; and for at least one row of sub-pixels of the plurality of sub-pixels, in the at least one light emission stage, a duration in which start time of an effective pulse of a second light emission control signal of a second light emission control signal line corresponding to the at least one row of sub-pixels lags behind start time of an effective pulse of a first light emission control signal of a first light emission control signal line corresponding to the at least one row of sub-pixels is delay time, wherein the delay time is  $\Delta t$ , and  $\Delta t \geq t_1$ ; and wherein the driving method comprises: applying a first light emission control signal to the first light emission control signal line; and applying a second light emission control signal to the second light emission control signal line, wherein in at least one light emission stage, start time of an effective pulse of the second light emission control signal lags behind start time of an effective pulse of the first light emission control signal.

19. The driving method according to claim 18, wherein the plurality of sub-pixels comprise a first color sub-pixel, a second color sub-pixel, and a third color sub-pixel, wherein a capacitance value of a pixel capacitor of a light-emitting element in the first color sub-pixel is greater than a capacitance value of a pixel capacitor of a light-emitting element in the second color sub-pixel, and the capacitance value of the pixel capacitor of the light-emitting element in the second color sub-pixel is greater than a capacitance value of a pixel capacitor of a light-emitting element in the third color sub-pixel; charging time of the pixel capacitor of the light-emitting element in the first color sub-pixel is  $t_2$ , wherein  $\Delta t \geq t_2$ .

20. The driving method according to claim 19, wherein delay time corresponding to each row of sub-pixels of the plurality of sub-pixels is equal.

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